ANALYTICALLY DENSE MORDELL–WEIL RANK JUMPS ON ELLIPTIC SURFACES VIA TRANSVERSE REALIZATION

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ABSTRACT. Let $\mathcal{W}_{\min,n}$ be the moduli stack of minimal Weierstrass fibrations of Faltings height n over an unparameterized $\mathbb{P}^1_{\mathbb{C}}$, and let $\mathcal{W}_n^{\mathrm{sb}}$ be its dense open substack of simply-branched Jacobian elliptic surfaces with only I_1 singular fibres. For integers $n \geq 2$ and r satisfying

$$1 \leq r \leq \left\lfloor \frac{10n-2}{n-1} \right\rfloor$$

we prove that every point of $\mathcal{W}_n^{\mathrm{sb}}$ is a limit of pairwise non-isomorphic surfaces in this locus carrying r independent Mordell–Weil sections, while the least canonical height of a non-zero section tends to infinity. Thus the rank-at-least-r locus is analytically dense in $\mathcal{W}_{\min,n}$. First- and second-order infinitesimal criteria also give exact-rank-r density for $1 \leq r \leq 10$ when $n \geq 3$, and for $1 \leq r \leq 18$ when $n = 2$. The direct realization argument combines Shepherd–Barron's infinitesimal period calculation with a block Vandermonde minor, lattice approximation, and the holomorphic implicit function theorem, and uses neither monodromy nor o-minimality. Lastly, we construct section moduli stacks of finite type parametrizing all-I_1 Jacobian elliptic surfaces of Faltings height n, equipped with r ordered non-zero sections of fixed canonical heights. At the constructed surfaces and sections, these algebraic stacks are smooth of dimension $10n-2-r(n-1)$, and their germs after base change to marked deformation charts identify with simultaneous fixed-class Hodge locus germs cut out transversely by the period equations.

1. INTRODUCTION

The global sections of a Jacobian elliptic surface form its Mordell–Weil group. At each fixed Faltings height $n \geq 2$, a very general such surface over $\mathbb{P}^1_{\mathbb{C}}$ has rank zero. For $n \geq 3$, this follows from Cox's result on the very general Picard number [Cox90, Prop. 3.2]; the elliptic $K3$ case $n = 2$ is classical. We study how additional sections appear under arbitrarily small complex deformations at the same height. We establish analytic density of Jacobian elliptic surfaces with Mordell–Weil rank at least r for every positive integer r in an explicit transverse range, and with rank exactly r for $1 \leq r \leq 10$ when $n \geq 3$ and $1 \leq r \leq 18$ when $n = 2$. Our construction coordinates local branch deformations through global period equations, producing transverse intersections of the elliptic surface period image with simultaneous Hodge loci. The resulting surfaces carry independent sections, while the least canonical height of a non-zero section tends to infinity. We also construct algebraic section moduli stacks of finite type parametrizing surfaces with ordered non-zero sections of fixed canonical heights. At the constructed tuples, their germs after base change to marked deformation charts identify with the corresponding smooth simultaneous fixed-class Hodge locus germs.

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Throughout the paper we work over $\mathbb{C}$. By a Jacobian elliptic surface over $\mathbb{P}^1$ we mean a smooth projective surface S equipped with a morphism $\pi\colon S \to \mathbb{P}^1$ with connected fibres and smooth generic fibre of genus one, together with a distinguished section $\sigma\colon \mathbb{P}^1 \to S$, called the zero section. We assume that π is relatively-minimal, meaning that no fibre contains a (-1)-curve. Write $\omega_{S/\mathbb{P}^1}$ for the relative dualizing sheaf. Its fundamental line bundle and Faltings height are

$$\mathcal{L} = (R^1\pi_*\mathcal{O}_S)^\vee \simeq \pi_*\omega_{S/\mathbb{P}^1}, \qquad n = \deg \mathcal{L} = \chi(\mathcal{O}_S)$$

We assume $n \geq 2$. The minimal discriminant divisor has degree $12n$; see [Mir89, BHPVdV04, SS10] for background on elliptic surfaces and their invariants.

Let $\mathcal{W}_{\min,n}$ denote the moduli stack of minimal Weierstrass fibrations of height n over an unparameterized $\mathbb{P}^1$. It is obtained from the fixed-base height moduli stack of minimal Weierstrass fibrations with smooth generic fibre by quotienting by changes-of-coordinates on the base [PS25, Thms. 1.1–1.2]. Since the base $\mathbb{P}^1$ is unparameterized, isomorphisms may change its coordinates but must carry fibres to fibres and preserve the zero section. The fixed-base height moduli description by minimal weighted linear series is given in [BPS24, Prop. 7.5]. We identify the geometric points of $\mathcal{W}_{\min,n}$ with isomorphism classes of relatively-minimal Jacobian elliptic surfaces by taking the minimal resolution of each Weierstrass model; conversely, the Weierstrass model is obtained by contracting the fibre components disjoint from the zero section [BPS24, Defs. 7.2–7.3]. Analytic density is understood in the usual complex analytic topology on the coarse moduli space; equivalently, it can be checked after pullback to an analytic atlas.

Write $E_S/\mathbb{C}(t)$ for the generic fibre and

$$\mathrm{MW}(S/\mathbb{P}^1) = E_S(\mathbb{C}(t))$$

for the Mordell–Weil group of sections. Let F and O denote the fibre and zero-section classes, and let $R(S)$ be the root lattice generated by the components of reducible fibres that do not meet O. Let $\mathrm{NS}(S)$ denote the Néron–Severi group of S. The trivial lattice $\mathrm{Triv}(S) \subset \mathrm{NS}(S)$ is generated by F, O, and these components. The Shioda–Tate description [Shi90, Thm. 1.3] gives

$$0 \longrightarrow \mathrm{Triv}(S) \longrightarrow \mathrm{NS}(S) \longrightarrow \mathrm{MW}(S/\mathbb{P}^1) \longrightarrow 0$$

Writing $\rho(S) = \mathrm{rk}\,\mathrm{NS}(S)$, we obtain

$$\rho(S) = 2 + \mathrm{rk}\,R(S) + \mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1) \tag{1}$$

The root-lattice term is determined by the Kodaira types of the singular fibres. To remove this contribution, we work on the squarefree-discriminant locus $\mathcal{W}_n^{\mathrm{I}_1} \subset \mathcal{W}_{\min,n}$. Every singular fibre on this locus is of Kodaira type I_1, that is, an irreducible rational curve with a single ordinary node. Thus $R(S) = 0$, and (1) becomes

$$\mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1) = \rho(S) - 2$$

On this locus, algebraic classes that are linearly independent modulo $\langle F, O\rangle$ give linearly independent Mordell–Weil sections.

To describe these classes, put

$$\Lambda_n = \langle F, O\rangle^\perp \subset H^2(S,\mathbb{Z})/\mathrm{tors}$$

Its isomorphism type depends only on n; we use the same symbol for a fixed abstract model. Throughout the paper, integral cohomology lattices are understood modulo torsion. For a lattice or finitely generated abelian group A, write $A_{\mathbf{K}} = A \otimes_{\mathbb{Z}} \mathbf{K}$ for $\mathbf{K} \in \{\mathbb{Q}, \mathbb{R}, \mathbb{C}\}$. For primitive $(1,1)$-cohomology, we write

$$H^{1,1}_{\mathrm{prim}}(S) = H^{1,1}(S) \cap \Lambda_{n,\mathbb{C}}$$
$$H^{1,1}_{\mathrm{prim}}(S,\mathbb{R}) = H^{1,1}(S) \cap \Lambda_{n,\mathbb{R}}$$

We use the half-normalized Néron–Tate pairing

$$\langle P, Q\rangle_{\widehat{h}} = \frac{1}{2}\langle P, Q\rangle_{\mathrm{Sh}}, \qquad \widehat{h}(P) = \langle P, P\rangle_{\widehat{h}}$$

where $\langle\,,\,\rangle_{\mathrm{Sh}}$ is Shioda's lattice height pairing. On the all-I_1 locus, a non-zero section P satisfies $\widehat{h}(P) = n + P \cdot O$, where $P \cdot O$ counts its intersections with the zero section, with multiplicity; see Proposition 3.22. On this locus, we call the orthogonal projection of the section class $[P]$ to Λ_n its *Shioda class*; explicitly, $\gamma_P = [P] - [O] - (n + P \cdot O)[F] \in \Lambda_n$. The Shioda map identifies the Mordell–Weil group with the primitive integral Hodge lattice on the all-I_1 locus:

$$\sigma_{\mathrm{Sh}} \colon \mathrm{MW}(S/\mathbb{P}^1) \xrightarrow{\sim} \Lambda_n \cap H^{1,1}(S), \qquad P \longmapsto \gamma_P$$

For this integral identification, see the proof of Theorem 3.11.

The moduli stack $\mathcal{W}_n^{\mathrm{I}_1}$ is irreducible of dimension $10n-2$, as follows from the quotient description in [PS25, Thms. 1.1 and 1.2(a)]. Each Jacobian elliptic surface in this locus has a classifying morphism [HP19, Prop. 9]

$$\varphi_S \colon \mathbb{P}^1 \longrightarrow \overline{\mathcal{M}}_{1,1}$$

Here $\overline{\mathcal{M}}_{1,1}$ is the moduli stack of stable genus one curves with one marked point, and $\mathcal{M}_{1,1}$ is its open substack parametrizing smooth curves. This morphism is unramified over the boundary $\overline{\mathcal{M}}_{1,1} \smallsetminus \mathcal{M}_{1,1}$; see the local calculation in the proof of [HP19, Prop. 11]. Its stack-theoretic ramification divisor $\mathrm{Ram}(\varphi_S)$ is the zero divisor of its differential; it differs from the ramification divisor of the coarse j-map. We call S *simply-branched* when $\mathrm{Ram}(\varphi_S)$ is reduced, and denote the resulting substack by

$$\mathcal{W}_n^{\mathrm{sb}} \subset \mathcal{W}_n^{\mathrm{I}_1}$$

Distinct ramification points are allowed to have the same image under φ_S. Lemmas 3.2 and 3.3 show that $\mathcal{W}_n^{\mathrm{sb}}$ is Zariski open and dense in $\mathcal{W}_{\min,n}$, hence also analytically dense.

Elliptic $K3$ surfaces with $n = 2$ have Kodaira dimension $\kappa = 0$. They provide the archetype for our question. Since $F \cdot F = 0$, $F \cdot O = 1$, and $O^2 = -2$, the classes $F, O + F$ identify $\langle F, O\rangle$ with the hyperbolic plane U. We recall a standard consequence of local Torelli and the Shioda–Tate formula, including its proof to explain the comparison with surfaces of height $n \geq 3$, which have Kodaira dimension $\kappa = 1$.

Proposition 1.1. *The locus*

$$\left\{[S] \in \mathcal{W}_2^{\mathrm{I}_1} : \mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1) = 18\right\}$$

is analytically dense in $\mathcal{W}_2^{\mathrm{I}_1}$. Consequently, for every $1 \leq r \leq 18$, the locus of Jacobian elliptic surfaces with Mordell–Weil rank at least r is analytically dense in $\mathcal{W}_2^{\mathrm{I}_1}$.

Proof. Fix $[S] \in \mathcal{W}_2^{\mathrm{I}_1}$. The lattice $\Lambda_2 = \langle F, O\rangle^\perp$ has signature $(2,18)$. By local Torelli, the marked deformations of S on which F and O remain algebraic are locally parametrized by an open subset of a connected component $\mathcal{D}_2^+$ of

$$\big\{\Pi \subset \Lambda_{2,\mathbb{R}} : \Pi \text{ is an oriented positive real two-plane}\big\}$$

see [Huy16, Ch. 6, Prop. 2.8 and §2.4]. The line bundles $\mathcal{O}_S(F)$ and $\mathcal{O}_S(O)$ extend over this fixed-class deformation space. On the central surface, their spaces of sections have dimensions two and one, respectively, and their first cohomology groups vanish. Cohomology and base change therefore extend the pencil and its zero-section divisor. After shrinking, the pencil remains basepoint free and the divisor remains a smooth rational curve; their intersection number one makes this divisor a section. The squarefree discriminant condition is open, so we obtain a marked chart B in $\mathcal{W}_2^{\mathrm{I}_1}$ whose period image is open in $\mathcal{D}_2^+$.

Positive two-planes defined over $\mathbb{Q}$ are dense in $\mathcal{D}_2^+$. Indeed, an oriented basis of any positive real two-plane can be approximated by vectors in $\Lambda_{2,\mathbb{Q}}$, and sufficiently close approximations still span positive planes with the induced orientation. Choose a rational plane Π in the period image of B, and let S_Π be the corresponding surface.

Since Π is defined over $\mathbb{Q}$, its orthogonal complement is a rational subspace of dimension 18. Hence

$$\operatorname{rk}\big(\Pi^\perp \cap \Lambda_2\big) = 18$$

These integral classes are of type $(1,1)$ on S_Π, so the Lefschetz $(1,1)$-theorem makes them algebraic. Together with F and O, these give 20 independent classes in $\mathrm{NS}(S_\Pi)$. Since divisor classes lie in $H^{1,1}(S_\Pi)$, we have

$$20 \le \rho(S_\Pi) \le h^{1,1}(S_\Pi) = 20$$

Hence $\rho(S_\Pi) = 20$.

The surface S_Π lies in $\mathcal{W}_2^{\mathrm{I}_1}$, so its singular fibres are irreducible. Thus (1) gives

$$\operatorname{rk}\mathrm{MW}(S_\Pi/\mathbb{P}^1) = \rho(S_\Pi) - 2 = 18$$

Rational positive planes occur in every open subset of the local period image, proving density. The assertion for rank at least r follows immediately. ∎

This is the classical period-theoretic density argument with the fibre and zero-section classes fixed. For related results, Oguiso proves density of every prescribed Picard number in the full Kuranishi space and constructs a hierarchy of narrow Mordell–Weil lattices of Jacobian $K3$ surfaces [Ogu03, Cor. 1.3 and Thm. 1.4].

In the $K3$ case, local Torelli identifies the marked deformation space preserving F and O with an open subset of $\mathcal{D}_2^+$. At higher height, the dimension of the ambient period domain prevents such local openness. The primitive Hodge numbers are

$$h^{2,0} = h^{0,2} = n-1, \qquad h^{1,1}_{\mathrm{prim}} = 10n-2$$

For general n, let $\mathcal{D}_n^+$ denote the connected component of the corresponding polarized weight-2 period domain containing the marked periods under consideration. For $n \ge 3$, its dimension satisfies

$$\dim_{\mathbb{C}} \mathcal{D}_n^+ = (n-1)(10n-2) + \binom{n-1}{2} > 10n-2 = \dim \mathcal{W}_n^{\mathrm{I}_1}$$

The local elliptic surface period image therefore has positive codimension in $\mathcal{D}_n^+$. Density of Hodge loci in the ambient domain alone does not guarantee that they meet this image.

A fixed class $\gamma \in \Lambda_n$ with $\gamma^2 < 0$ defines a Hodge locus of complex codimension $n-1$ in $\mathcal{D}_n^+$. Such ambient codimension calculations do not by themselves establish independence of the period differentials after restriction to the elliptic surface deformation space. Simultaneous Hodge conditions can also give intersections of dimension greater than expected.

For an ordered tuple $\mathbf{d} = (d_1, \ldots, d_r)$ of positive integers, let $\mathcal{S}_{n,\mathbf{d}}^{\mathrm{I}_1}$ denote the section moduli stack of height n all-I_1 Jacobian elliptic surfaces equipped with ordered non-zero sections $P_1, \ldots, P_r$ satisfying $\widehat{h}(P_a) = d_a$. The base is considered up to isomorphism. Subsection 3.8 constructs this algebraic Deligne–Mumford stack of finite type and its representable forgetful morphism

$$\rho : \mathcal{S}_{n,\mathbf{d}}^{\mathrm{I}_1} \longrightarrow \mathcal{W}_n^{\mathrm{I}_1}$$

The canonical heights are locally constant in families with chosen sections, so fixing $\mathbf{d}$ does not change the deformation germ of a tuple.

Our main theorem constructs independent Mordell–Weil sections through transverse period conditions, with the least non-zero section height tending to infinity while the Faltings height remains fixed.

Main Theorem 1.2. *Let $n \geq 2$ and $r \geq 1$ be integers satisfying*

$$r(n-1) \leq 10n - 2$$

Let $\mathcal{W}_{\min,n}$ denote the complex moduli stack of minimal Weierstrass fibrations of Faltings height n over an unparameterized $\mathbb{P}^1_{\mathbb{C}}$. Let $\mathcal{W}_n^{\mathrm{sb}} \subset \mathcal{W}_{\min,n}$ be the locus where every singular fibre has Kodaira type I_1 and the differential of the classifying morphism

$$\varphi_S : \mathbb{P}^1_{\mathbb{C}} \longrightarrow \overline{\mathcal{M}}_{1,1}$$

has reduced zero divisor. Every $[S] \in \mathcal{W}_n^{\mathrm{sb}}$ is an analytic limit of pairwise non-isomorphic surfaces $[S_m] \in \mathcal{W}_n^{\mathrm{sb}}$ carrying r independent Mordell–Weil sections $P_{1,m}, \ldots, P_{r,m}$ with the following properties.

(1) *The Shioda classes $\gamma_{1,m}, \ldots, \gamma_{r,m} \in \Lambda_n$ of the chosen sections are indivisible, and*

$$\min_{0 \neq P \in \mathrm{MW}(S_m/\mathbb{P}^1)} \widehat{h}(P) \longrightarrow \infty \qquad \text{as } m \to \infty$$

(2) *On a marked local deformation chart, the $r(n-1)$ simultaneous fixed-class period equations for these classes have linearly independent differentials at S_m. Their common zero locus is therefore smooth of complex codimension $r(n-1)$ near S_m.*

(3) *Put $\mathbf{d}_m = (\widehat{h}(P_{1,m}), \ldots, \widehat{h}(P_{r,m}))$. The stack $\mathcal{S}_{n,\mathbf{d}_m}^{\mathrm{I}_1}$ is smooth at $(S_m, P_{1,m}, \ldots, P_{r,m})$ of complex dimension $10n - 2 - r(n-1)$. After base change to the chart in* (2)*, the forgetful morphism identifies its germ at the chosen lift with the germ of the common zero locus in* (2)*. On a sufficiently small connected representative, the chosen sections extend uniquely and their height-pairing matrix is constant.*

Consequently, the rank-at-least-r locus in $\mathcal{W}_n^{\mathrm{sb}}$ is analytically dense in $\mathcal{W}_{\min,n}$. Taking $r = 10$ gives 8-dimensional simultaneous Hodge germs for every $n \geq 2$. Their representatives can be chosen with images in any prescribed non-empty analytic open subset and with arbitrarily large uniform lower bounds for all non-zero section heights; see Corollary 3.13.

We record the period equations on a single marked deformation chart.

Theorem 1.3. *The surfaces and sections in Theorem 1.2 may be chosen in a single simply-connected analytic étale local deformation chart $B \to \mathcal{W}_n^{\mathrm{sb}}$ of dimension $10n-2$, with $S_{b_0} = S$, $S_{b_m} = S_m$, and $b_m \to b_0$. A marking identifies the lattices*

$$\langle F_b, O_b \rangle^{\perp} \subset H^2(S_b, \mathbb{Z})/\mathrm{tors}$$

with a fixed lattice Λ_n by parallel transport preserving the cup-product pairing, where F_b, O_b are the fibre and zero-section classes.

After shrinking B and discarding finitely many initial terms, choose a holomorphic frame $\omega_0, \dots, \omega_{n-2}$ of F^2. For the Shioda classes $\gamma_{1,m}, \dots, \gamma_{r,m}$ of the chosen sections, define

$$\Phi_m \colon B \longrightarrow \mathbb{C}^{r(n-1)} \qquad \Phi_m(b) = \big(\gamma_{a,m} \cdot \omega_j(b)\big)_{\substack{1 \leq a \leq r \\ 0 \leq j \leq n-2}}$$

Then

$$\Phi_m(b_m) = 0 \qquad \mathrm{rk}_{\mathbb{C}}(d\Phi_m)_{b_m} = r(n-1)$$

After base change to B, forgetting the chosen ordered sections identifies the section-moduli germ at the chosen lift of b_m with the Hodge germ $(\Phi_m^{-1}(0), b_m)$.

Schiffer variations arise from elementary local changes of the holomorphic gluing of a Riemann surface [Sch53, §3]. On the simply-branched locus, Shepherd–Barron's adaptation to elliptic surfaces provides local deformation coordinates indexed by the ramification points of the classifying morphism [SB23, Prop. 4.4]. Their tangent directions form a basis of the deformation tangent space. Each direction induces a rank-one map from $H^{2,0}(S)$ to $H^{1,1}_{\mathrm{prim}}(S)$ [SB23, Thm. 1.4(3)]. We call non-zero generators of these image lines the *Schiffer image classes*; they form an orthogonal basis of primitive $(1,1)$-cohomology [SB23, Thm. 4.11(4)]. Evaluation of holomorphic two-forms at distinct ramification points gives Vandermonde minors. Our block construction uses these to show that a maximal Jacobian minor of the simultaneous period equations is not identically zero.

We choose a real tuple in $H^{1,1}_{\mathrm{prim}}(S, \mathbb{R})$ with negative-definite Gram matrix at which the chosen minor does not vanish. We approximate its directions by those of indivisible lattice vectors whose norms tend to infinity in a fixed auxiliary Euclidean norm. Applying the holomorphic implicit function theorem, with the normalized lattice vectors as parameters, gives nearby surfaces on which all their period pairings vanish. Rescaling to the original integral vectors preserves these vanishing pairings. Since the vectors are real, they are also orthogonal to $H^{0,2}$ and hence have type $(1,1)$.

Independence persists under sufficiently close approximation. Moreover, the normalized limiting vectors have negative square, so $-\gamma_{a,m}^2 \to +\infty$ for every a. The Lefschetz $(1,1)$-theorem makes these classes algebraic, and on the all-I_1 locus

Shioda's identification turns them into independent sections satisfying

$$\widehat{h}(P_{a,m}) = -\frac{1}{2}\gamma_{a,m}^2 \longrightarrow +\infty$$

These heights are integers because Λ_n is even. To control the heights of all non-zero sections, we use Lemma 3.7. On a relatively compact marked chart, only finitely many non-zero integral classes can become Hodge with $-\gamma^2/2 \leq H$. Their proper closed Hodge loci have an open dense complement. Applying the local realization theorem in these complements, with increasing H and shrinking neighbourhoods, makes the least canonical height of a non-zero section tend to infinity.

To obtain exact rank 1, we use an explicit infinitesimal criterion. Suppose that a non-zero class $\alpha \in \Lambda_n$ is indivisible, is of type $(1,1)$ at a point of $\mathcal{W}_n^{\mathrm{sb}}$, and pairs non-trivially with every Schiffer image class. For each $\beta \in \Lambda_n \smallsetminus \mathbb{Q}\alpha$ that is Hodge at that point, some first-order deformation preserving the Hodge type of α does not preserve that of β. Consequently, outside a countable union of proper closed analytic subsets, a sufficiently small connected representative of the 1-class Hodge germ parametrizes surfaces of Mordell–Weil rank exactly 1; see Proposition 3.16. This criterion and its simultaneous version, combined with rational approximation and local realization, give exact-rank density for $r \leq 9$ when $n \geq 3$ and for $r \leq 18$ when $n = 2$. The second-order argument of Subsection 3.7 supplies the remaining case $r = 10$ when $n \geq 3$. Together these arguments give the following result, proved in Theorem 3.14.

Corollary 1.4. *Let $n \geq 2$ and $r \geq 1$ be integers, with $r \leq 18$ if $n = 2$ and $r \leq 10$ if $n \geq 3$. Then the locus of surfaces in $\mathcal{W}_n^{\mathrm{sb}}$ with Mordell–Weil rank exactly r is analytically dense in $\mathcal{W}_{\min,n}$.*

These first- and second-order arguments use neither monodromy nor o-minimality.

The constant dimension 8 also has a natural moduli-theoretic interpretation: 10 additional sections together with the zero section give 11 markings, and the corresponding 11-dimensional fibre-product target has trivial canonical bundle on the locus containing the section maps. Remark 3.25 explains the expected deformation dimension $11 - 3 = 8$ for these maps from $\mathbb{P}^1$, after quotienting by source automorphisms. It also compares this threshold with the transition in $\kappa(\overline{M}_{1,r+1})$ from $-\infty$ for $1 \leq r \leq 9$ to 0 at $r = 10$ and then to 1 for $r \geq 11$.

Cox and Kloosterman studied the Noether–Lefschetz and Mordell–Weil rank loci of Jacobian elliptic surfaces. For $n \geq 3$, Cox proves that the positive Mordell–Weil rank locus has dimension $9n - 1$ [Cox90] and that the union of the codimension-$(n-1)$ components of the Noether–Lefschetz locus is dense in the classical topology [Cox90, Thm. 3.1(iii)]. For every $n \geq 2$ and every $H > 0$, he also proves density of surfaces with positive Mordell–Weil rank for which every non-zero section P satisfies $P \cdot O \geq H$ [Cox90, Thm. 4.4]. On the all-I_1 locus, our normalization satisfies $\widehat{h}(P) = n + P \cdot O$, so this also gives arbitrarily large lower bounds for all non-zero section heights. Kloosterman determines the dimensions of higher Noether–Lefschetz loci on the nonconstant-j locus and obtains upper bounds for the dimensions of Mordell–Weil rank loci there [Klo07, Thm. 1.1 and Cor. 1.3].

These results fit into the general study of density and distribution of Hodge loci for variations of Hodge structure. Khelifa–Urbanik give density criteria under monodromy and expected-dimension hypotheses [KU25]. Under suitable hypotheses on a weight-two variation, Tayou–Tholozan prove equidistribution of transverse refined Noether–Lefschetz points, ordered by lattice discriminant, when the expected dimension is zero [TT23, Thm. 6.10].

Remark 1.5. Higher Mordell–Weil ranks occur in classical Delsarte examples with cyclic symmetry; see [Hei12] for a classification and rank computations. Shioda's isotrivial example has rank 68 at height 60 [Shi92, pp. 109–110]. The non-isotrivial example studied by Shioda [Shi86] and Stiller [Sti87, Thm. 4.2] has rank 56 at height 210. The former has only type-II singular fibres. The latter admits the model

$$y^2 = x^3 + x + t^{1260}$$

whose singular fibres are all of type I_1, but whose stack-theoretic ramification divisor is

$$\mathrm{Ram}(\varphi_S) = 1259[0] + 839[\infty]$$

Thus neither example lies in $\mathcal{W}_n^{\mathrm{sb}}$. Nevertheless, these classical examples can be approximated, at fixed Faltings height, by surfaces in our controlled rank-10 families; see Theorem 3.14.

The natural question of whether Mordell–Weil ranks of positive-height Jacobian elliptic surfaces over $\mathbb{P}^1_{\mathbb{C}}$ are unbounded as the height grows remains open [SS10, Question 14.3]. Over each fixed finite field $\mathbb{F}_q$, Mordell–Weil ranks over $\mathbb{F}_q(t)$ are unbounded: Tate–Shafarevich established this for isotrivial curves in odd characteristic [Tv67], and Ulmer constructed non-isotrivial examples in every positive characteristic [Ulm02, Thm. 1.5].

1.1. **Outline of the paper.** Section 2 records the primitive lattice and the period conditions used in the proof. Section 3 establishes simultaneous transverse realization, the height bounds, and analytic density. Subsections 3.6 and 3.7 establish exact-rank density in the ranges of Theorem 3.14 using first- and second-order period equations. Finally, Subsection 3.8 constructs the section moduli stacks, identifies canonical heights with two-pointed degrees, and proves the section moduli smoothness and germ comparison, completing the proofs of Theorems 1.2 and 1.3.

2. Primitive cohomology and period conditions

We recall the lattice and Hodge-theoretic facts used in the realization argument. Our general reference for Hodge theory, period domains, and variations of Hodge structure throughout the paper is Carlson–Müller-Stach–Peters [CMSP17]. We use a dot for the intersection pairing and its complex-bilinear extension.

2.1. **The primitive lattice.** Let S be a smooth complex projective surface, and let $\pi\colon S \to \mathbb{P}^1$ be a relatively-minimal Jacobian elliptic fibration with $\chi(\mathcal{O}_S) = n \geq 2$. Let F and O denote a fibre divisor and the zero section, respectively; we use the same symbols for their divisor classes. The canonical-bundle formula gives $K_S =$

$(n-2)F$. Since $F \cdot F = 0$ and $F \cdot O = 1$, adjunction gives $O^2 = -n$. The standard invariant formulas

$$q(S) = 0, \qquad p_g(S) = n-1, \qquad e(S) = 12n$$

give $b_2(S) = 12n-2$; see [Mir89, SS10]. Hodge decomposition and the Hodge index theorem give intersection signature $(2n-1, 10n-1)$ on $H^2(S,\mathbb{Z})/\mathrm{tors}$.

The fibre and zero-section classes have intersection matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & -n \end{pmatrix}$$

Its determinant is -1, so $\langle F, O\rangle$ is a primitive unimodular sublattice of signature $(1,1)$. Its orthogonal complement

$$\Lambda_n = \langle F, O\rangle^{\perp} \subset H^2(S,\mathbb{Z})/\mathrm{tors}$$

therefore has rank $12n-4$ and signature $(2n-2, 10n-2)$. Orthogonal projection gives an integral decomposition

$$H^2(S,\mathbb{Z})/\mathrm{tors} = \langle F, O\rangle \oplus \Lambda_n$$

We recall the standard description of the primitive lattice and its proof [Gre20, Thm. 24]. The lattice Λ_n is even. Indeed, the Wu formula gives

$$\gamma^2 \equiv \gamma \cdot K_S = (n-2)\gamma \cdot F = 0 \pmod 2 \qquad (\gamma \in \Lambda_n)$$

Poincaré duality makes $H^2(S,\mathbb{Z})/\mathrm{tors}$ unimodular, so the orthogonal decomposition also shows that Λ_n is unimodular. The classification of indefinite even unimodular lattices gives

$$\Lambda_n \simeq U^{\oplus(2n-2)} \oplus E_8(-1)^{\oplus n} \tag{2}$$

Here U is the even hyperbolic plane, with Gram matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and $E_8(-1)$ is the negative-definite E_8 lattice; see [Ser73, Ch. V, Thm. 5]. We use Λ_n also for a fixed abstract model of this lattice.

The primitive Hodge numbers are $h^{2,0} = h^{0,2} = p_g(S) = n-1$ and $h^{1,1}_{\mathrm{prim}} = 10n-2$. The intersection pairing is positive definite on the real subspace

$$\left(H^{2,0}(S) \oplus H^{0,2}(S)\right) \cap \Lambda_{n,\mathbb{R}}$$

and negative definite on $H^{1,1}_{\mathrm{prim}}(S,\mathbb{R})$. In particular, every non-zero class $\gamma \in \Lambda_n \cap H^{1,1}(S)$ satisfies

$$-\frac{1}{2}\gamma^2 \in \mathbb{Z}_{>0}$$

Here primitive cohomology means the orthogonal complement of F and O. To avoid a clash with this usage, we call a lattice vector *indivisible* if it is non-zero and not of the form $k\gamma'$ with γ' in the lattice and $k \geq 2$ an integer.

2.2. **Periods and fixed-class Hodge loci.** A primitive Hodge structure is determined by its space of holomorphic two-forms. Let $\mathcal{D}_n^+$ be a connected component of the period domain parametrizing $(n-1)$-dimensional subspaces $F^2 \subset \Lambda_{n,\mathbb{C}}$ satisfying

$$F^2 \cdot F^2 = 0, \qquad \omega \cdot \overline{\omega} > 0 \quad \text{for every } 0 \neq \omega \in F^2$$

Its complex dimension is $(n-1)(10n-2)+\binom{n-1}{2}$. It is a type-IV Hermitian symmetric domain when $n = 2$; for $n \geq 3$, it is a Griffiths period domain that is not Hermitian symmetric. For this period-domain description in the elliptic-surface setting, see [Gre20, §3, p. 2345]; for the general Hodge-theoretic background, see [CMSP17, Ch. 4].

For a real class $\gamma \in \Lambda_{n,\mathbb{R}}$, the condition

$$\gamma \perp F^2$$

is equivalent to being of type $(1,1)$. Indeed, reality also gives orthogonality to $\overline{F^2}$, and the orthogonal complement of $F^2 \oplus \overline{F^2}$ is the primitive $(1,1)$-subspace.

If $\gamma^2 < 0$, its fixed-class Hodge locus in $\mathcal{D}_n^+$ is smooth of complex codimension $n-1$. Requiring $F^2 \subset \gamma^\perp$ removes one negative direction from the intersection space, replacing $10n-2$ by $10n-3$ in the dimension formula above. More generally, a fixed negative-definite span of dimension r defines a smooth ambient Hodge locus of complex codimension $r(n-1)$.

To express these conditions on a family, let B be a small simply-connected chart carrying a smooth projective family of height n Jacobian elliptic surfaces with zero section. The fibre and zero-section classes remain flat and algebraic. Choose a marking at a base point $b_0 \in B$, that is, an isometry from the primitive cohomology lattice of S_{b_0} to Λ_n. Since B is simply-connected, parallel transport extends this to compatible identifications of the primitive cohomology lattices of all fibres with Λ_n. Choose a local holomorphic frame $\omega_0(b), \dots, \omega_{n-2}(b)$ of the Hodge bundle F^2. For fixed classes $\gamma_1, \dots, \gamma_r \in \Lambda_{n,\mathbb{R}}$, their simultaneous Hodge locus is defined by the holomorphic equations

$$\gamma_a \cdot \omega_j(b) = 0, \qquad 1 \leq a \leq r, \quad 0 \leq j \leq n-2$$

Let $\mathrm{per}\colon B \to \mathcal{D}_n^+$ be the marked period map, and let $\mathcal{D}_\gamma \subset \mathcal{D}_n^+$ be the locus where all the classes γ_a have type $(1,1)$. Their simultaneous Hodge locus Z_γ in B fits into the Cartesian square

$$\begin{array}{ccc} Z_\gamma & \hookrightarrow & B \\ \downarrow & & \downarrow{\scriptstyle \mathrm{per}} \\ \mathcal{D}_\gamma & \hookrightarrow & \mathcal{D}_n^+ \end{array}$$

These are $r(n-1)$ equations, but their differentials need not be independent on an elliptic surface deformation space. Section 3 constructs suitable integral classes and nearby surfaces at which all these differentials are independent.

3. Geometric realization and density of Mordell–Weil rank jumps

We construct transverse intersections of the local period image of $\mathcal{W}_n^{\mathrm{sb}}$ with simultaneous fixed-class Hodge loci in $\mathcal{D}_n^+$. We then identify the resulting integral Hodge classes with independent Mordell–Weil sections. Fix a height $n \geq 2$, and put $h := n-1$ and $N := 10n-2$.

The all-I_1 locus $\mathcal{W}_n^{\mathrm{I}_1}$ is obtained from a non-empty Zariski-open subset of the coefficient space

$$V_n := H^0(\mathbb{P}^1, \mathcal{O}(4n)) \oplus H^0(\mathbb{P}^1, \mathcal{O}(6n))$$

by quotienting by Weierstrass scaling and changes of coordinates on the base [PS25, Thms. 1.1 and 1.2(a)]. The coefficient space is irreducible, so its non-empty Zariski-open subset and the quotient $\mathcal{W}_n^{\mathrm{I}_1}$ are irreducible. Moreover,

$$\dim \mathcal{W}_n^{\mathrm{I}_1} = (4n+1) + (6n+1) - 1 - 3 = 10n-2 = N$$

We use the primitive lattice and period equations of Section 2. The construction applies to every positive integer r satisfying $rh \leq N$; neither the integral classes nor their Gram matrices are prescribed in advance.

3.1. **The simply-branched locus.**

We first identify a dense open locus on which each ramification point supplies one deformation parameter. Let

$$\varphi_S \colon \mathbb{P}^1 \longrightarrow \overline{\mathcal{M}}_{1,1} \simeq \mathcal{P}(4,6)$$

be the classifying morphism of $[S] \in \mathcal{W}_n^{\mathrm{I}_1}$, where $\mathcal{P}(4,6)$ is the weighted projective stack. Following [SB23, §2], we write $\mathrm{Ram}(\varphi_S)$ for the zero divisor of the stack-theoretic differential

$$\mathrm{d}\varphi_S \colon T_{\mathbb{P}^1} \longrightarrow \varphi_S^* T_{\overline{\mathcal{M}}_{1,1}}$$

This differs from the ramification divisor of the coarse j-map at the orbifold values $j = 0, 1728$. The all-I_1 condition makes φ_S non-constant and unramified over the boundary.

By [SB23, Lem. 2.1 and Prop. 2.8(4)], we have $T_{\overline{\mathcal{M}}_{1,1}} \simeq \mathcal{O}(10)$ and $\varphi_S^*\mathcal{O}(1) \simeq \mathcal{O}_{\mathbb{P}^1}(n)$. Thus $\mathrm{d}\varphi_S$ is a non-zero section of $\mathcal{O}_{\mathbb{P}^1}(10n-2)$, and

$$\deg \mathrm{Ram}(\varphi_S) = 10n-2 = N$$

For Weierstrass coefficients $a_i \in H^0(\mathbb{P}^1, \mathcal{O}(in))$, $i = 4, 6$, this divisor is cut out, including multiplicities, by the weighted Wronskian

$$\Theta(a_4, a_6) := 3a_6\,\mathrm{d}a_4 - 2a_4\,\mathrm{d}a_6 \in H^0\big(\mathbb{P}^1, \mathcal{O}(10n) \otimes \Omega^1_{\mathbb{P}^1}\big)$$

Here the differentials are computed in local frames of $\mathcal{O}(n)$. Under a change of frame by a unit u, the terms involving $\mathrm{d}u$ cancel, giving $\Theta(u^{-4}a_4, u^{-6}a_6) = u^{-10}\Theta(a_4, a_6)$; hence the expression defines the stated global section. This agrees, up to a non-zero constant, with Shepherd–Barron's expression for the ramification divisor [SB23, §10, Lem. 10.2]. Since a_4 and a_6 have no common zero, the infinitesimal scaling vector $(4a_4, 6a_6)$ is nowhere zero. Taking its determinant with $(\mathrm{d}a_4, \mathrm{d}a_6)$ represents the differential by $4a_4\,\mathrm{d}a_6 - 6a_6\,\mathrm{d}a_4 = -2\Theta(a_4, a_6)$, so the zero divisors agree, including multiplicities.

Definition 3.1. A surface $[S] \in \mathcal{W}_n^{\mathrm{I}_1}$ is *simply-branched* if the stack-theoretic ramification divisor $\mathrm{Ram}(\varphi_S)$ is reduced. We denote the corresponding substack by $\mathcal{W}_n^{\mathrm{sb}} \subset \mathcal{W}_n^{\mathrm{I}_1}$. Thus, after passing to an étale neighbourhood which orders the ramification divisor, a point of $\mathcal{W}_n^{\mathrm{sb}}$ has

$$\mathrm{Ram}(\varphi_S) = p_1 + \cdots + p_N$$

with the p_i pairwise distinct. This definition does not require the images $\varphi_S(p_i)$ to be pairwise distinct.

Lemma 3.2. *The substack $\mathcal{W}_n^{\mathrm{sb}}$ is Zariski open and dense in $\mathcal{W}_n^{\mathrm{I}_1}$.*

Proof. Reducedness of the zero divisor of $\Theta(a_4, a_6)$ is a Zariski-open condition on the Weierstrass coefficients. Thus $\mathcal{W}_n^{\mathrm{sb}}$ is Zariski open in $\mathcal{W}_n^{\mathrm{I}_1}$. To prove density, we show that multiple ramification points can be split by deforming the surface.

We use Shepherd–Barron's local deformation theory. The height n, base-genus-zero part of his stack $\mathcal{JE}^{\mathrm{gen}}$ is our all-I_1 locus [SB23, p. 2523]. Its classifying morphisms are non-constant and unramified over the boundary (see the local calculation in the proof of [HP19, Prop. 11]); their ramification divisors need not be reduced. His numerical hypotheses hold because the base has genus zero and $p_g = n - 1 > 0$. Moreover, the relatively-minimal surface coincides with its minimal Weierstrass model, since there are no fibre components disjoint from the zero section.

Here $\mathrm{Def}(S)$ denotes the miniversal analytic deformation space of the Jacobian elliptic fibration with its zero section, with the base considered up to isomorphism. It underlies a local quotient chart of $\mathcal{W}_n^{\mathrm{I}_1}$ at $[S]$ [SB23, Prop. 2.9(2)–(3) and Prop. 4.4(1)].

Suppose that the ramification divisor of a given surface has multiplicity $e_i - 1$ at a point p_i. Shepherd–Barron's local isomorphism [SB23, Prop. 4.4] identifies this miniversal chart with the product of the deformation spaces of the finite schemes $\operatorname{Spec} \mathbb{C}[t]/(t^{e_i})$. The i-th local deformation base has dimension $e_i - 1$. Since $\sum_i (e_i - 1) = \deg \mathrm{Ram}(\varphi_S) = N$, their product is an N-dimensional miniversal chart. On the simply-branched locus, each factor is one-dimensional, giving one deformation parameter for each ramification point.

More generally, the product description allows us to deform the ramification at the different points independently. To split a point of multiplicity $e - 1$, lift φ_S locally to an étale chart of the target stack and choose coordinates w on the source and s on that chart. Shepherd–Barron's Schiffer deformation has the form [SB23, Prop. 4.3]

$$s = w^e + e \sum_{j=0}^{e-2} t_{e-j} w^j$$

Up to the non-zero factor e, its critical divisor is therefore cut out by

$$w^{e-1} + \sum_{j=1}^{e-2} j\, t_{e-j} w^{j-1}$$

For $e = 2$ this divisor is already reduced. For $e \geq 3$, setting $t_{e-1} \neq 0$ and all other relevant parameters equal to zero gives the polynomial $w^{e-1} + t_{e-1}$, which

has distinct roots. Thus the discriminant of the critical polynomial, viewed as a function of the deformation parameters, is not identically zero.

Choose pairwise disjoint coordinate discs around the points p_i. By Rouché's theorem, after shrinking the deformation chart each disc contains exactly $e_i - 1$ critical points, counted with multiplicity. These account for the total ramification degree N, so there are no critical points outside their union. The simultaneous non-vanishing of the local discriminants defines a dense open subset of the miniversal chart. Thus every surface in $\mathcal{W}_n^{\mathrm{I}_1}$ is an analytic limit of surfaces with reduced ramification.

Since $\mathcal{W}_n^{\mathrm{I}_1}$ is irreducible by the quotient description in [PS25, Thms. 1.1 and 1.2(a)], its non-empty Zariski-open substack $\mathcal{W}_n^{\mathrm{sb}}$ is Zariski dense. ∎

Lemma 3.3. *For every $n \geq 2$, both inclusions*

$$\mathcal{W}_n^{\mathrm{sb}} \subset \mathcal{W}_n^{\mathrm{I}_1} \subset \mathcal{W}_{\min,n}$$

are Zariski open and dense. In particular, $\mathcal{W}_n^{\mathrm{sb}}$ is analytically dense in $\mathcal{W}_{\min,n}$.

Proof. Minimality is a Zariski-open condition on the Weierstrass data in the irreducible coefficient space V_n [BPS24, Prop. 4.15]. Intersecting with the open condition $\Delta \not\equiv 0$ gives a non-empty Zariski-open subset of V_n presenting $\mathcal{W}_{\min,n}$ after quotienting by Weierstrass scaling and changes of coordinates on the base [PS25, Thms. 1.1–1.2].

Reducedness of the discriminant divisor on $\mathbb{P}^1$ is also a Zariski-open condition. This condition is non-empty: in homogeneous coordinates $[s : t]$ on $\mathbb{P}^1$, take $a_4 = -3s^{4n}$ and $a_6 = t^{6n}$. Then

$$\Delta = -16(4a_4^3 + 27a_6^2) = -432(t^{12n} - 4s^{12n})$$

has $12n$ distinct zeros, and it does not vanish at $[s : t] = [0 : 1]$. These Weierstrass data are minimal. Indeed, non-minimality at a point would force the discriminant to vanish there to order at least 12.

Thus the squarefree-discriminant locus is a non-empty Zariski-open subset of the irreducible minimal elliptic coefficient locus. Passing to the quotient shows that $\mathcal{W}_n^{\mathrm{I}_1}$ is Zariski open and dense in $\mathcal{W}_{\min,n}$.

Lemma 3.2 gives the same conclusion for $\mathcal{W}_n^{\mathrm{sb}} \subset \mathcal{W}_n^{\mathrm{I}_1}$. Hence $\mathcal{W}_n^{\mathrm{sb}}$ is Zariski open and dense in $\mathcal{W}_{\min,n}$. A non-empty Zariski-open subset of an irreducible complex variety is analytically dense; applying this on the coefficient presentation gives the analytic assertion. ∎

3.2. **The Shepherd–Barron frame.** We now choose coordinates and a cohomology basis in which the first derivative of the period map is explicit. This will let us test transversality of the period equations and show that bounded-height Hodge classes can be avoided locally.

Fix $[S] \in \mathcal{W}_n^{\mathrm{sb}}$ and order its ramification points as $p_1, \ldots, p_N$. Choose an affine coordinate z on the base for which none of the p_i lies at infinity, and put $z_i = z(p_i)$. On the corresponding affine Weierstrass chart, write

$$y^2 = x^3 + a_4(z)x + a_6(z) \qquad \omega_0 = \frac{\mathrm{d}x \wedge \mathrm{d}z}{2y}$$

The canonical bundle formula identifies global holomorphic two-forms with polynomials in z of degree at most $n-2$, multiplied by ω_0. Thus

$$H^{2,0}(S) = \bigoplus_{j=0}^{h-1} \mathbb{C}\,\omega_j \qquad \omega_j = z^j \omega_0$$

The form ω_0 is nowhere vanishing over the affine z-line. We use this polynomial identification for sections of $\mathcal{O}_{\mathbb{P}^1}(n-2)$ below; in particular, $f(p_i)$ means the scalar $f(z_i)$.

Every automorphism of the Jacobian fibration preserves its zero section and the reduced discriminant divisor of degree $12n$. Its induced automorphism of $\mathbb{P}^1$ therefore belongs to the finite group preserving these $12n \geq 3$ points, while the kernel embeds in the finite group $\mathrm{Aut}_{\mathbb{C}(t)}(E_S, 0)$. Thus $G \coloneqq \mathrm{Aut}_{\mathcal{W}_n^{\mathrm{sb}}}([S])$ is finite. Since $\mathcal{W}_n^{\mathrm{sb}}$ is an open substack of $\mathcal{W}_{\min,n}$, it is smooth and separated Deligne–Mumford by [PS25, Thm. 1.2(a)–(b)]. By [ES18, Prop. 2.9], it has an analytic open neighbourhood of the form $[Z/K]$ with K finite. Restricting near a lift of $[S]$ to its stabilizer G and linearizing the finite group action gives an analytic open immersion

$$[B/G] \longrightarrow \mathcal{W}_n^{\mathrm{sb}}$$

where B is a simply-connected analytic ball and $0 \in B$ lifts $[S]$ and is fixed by G. The quotient B/G identifies with an analytic neighbourhood of the image of $[S]$ in the coarse moduli space.

The relative ramification divisor is finite étale of degree N over B. Since B is simply-connected, the ordering at 0 extends to an ordering throughout B. We trivialize the primitive integral local system over B by flat transport. By the preceding deformation-space identification and [SB23, §4, Prop. 4.4], after shrinking B there are analytic coordinates $(t_1, \ldots, t_N)$ such that $\partial/\partial t_i|_0$ spans the Schiffer direction associated with p_i.

Shepherd–Barron associates to each p_i a meromorphic two-form $\widetilde{\eta}_i$ with poles of order at most two along the fibre over p_i and no other poles. It is of the second kind, meaning that its residue along that fibre vanishes. Its class is taken in the primitive quotient $F^{\perp}/\mathbb{C}F$. The zero section identifies this quotient with $\Lambda_{n,\mathbb{C}}$ by

$$[x] \longmapsto x - (x \cdot O)F$$

This expression is independent of the representative $x \in F^{\perp}$ and is orthogonal to both F and O. Let $\eta_i = \mathrm{pr}^{1,1}[\widetilde{\eta}_i] \in H^{1,1}_{\mathrm{prim}}(S)$ denote its $(1,1)$-component.

Shepherd–Barron's meromorphic classes $[\widetilde{\eta}_i]$ lie in $H^{2,0}(S) \oplus H^{1,1}_{\mathrm{prim}}(S)$ [SB23, Thm. 4.10]. The $H^{2,0}(S)$-summand pairs trivially with this space, so $[\widetilde{\eta}_i] \cdot [\widetilde{\eta}_j] = \eta_i \cdot \eta_j$. Since the ramification divisor is reduced, the projected classes $\eta_1, \ldots, \eta_N$ form a basis of $H^{1,1}_{\mathrm{prim}}(S)$ orthogonal for the complex-bilinear intersection pairing [SB23, Thm. 4.11(4)].

Write $E_{p_i} = \pi^{-1}(p_i)$ for the fibre over p_i. Let F^2 denote the holomorphic Hodge bundle over B, whose fibre at b is $H^{2,0}(S_b)$. The Gauss–Manin connection ∇ differentiates sections of the cohomology bundle using its flat trivialization. For every $\omega \in H^{2,0}(S)$, Shepherd–Barron's rank-one formula [SB23, Thm. 1.4(3)] gives

$$\left(\nabla_{\partial/\partial t_i}\omega\right)^{1,1} = c_i(\omega)\eta_i$$

The derivative is evaluated at $0 \in B$ using any local holomorphic extension of ω in F^2. Its $(1,1)$-projection is independent of that extension: changing the extension only adds an element of $H^{2,0}(S)$ to the derivative; see [CMSP17, Lem. 5.3.2]. Here c_i is a linear functional with

$$\ker(c_i) = H^0(S, \Omega_S^2(-E_{p_i}))$$

Since $K_S \simeq \pi^*\mathcal{O}_{\mathbb{P}^1}(n-2)$, the canonical linear system is base-point free for $n \geq 2$. Under the polynomial identification above, vanishing along E_{p_i} is the single condition $f(p_i) = 0$. Hence this kernel is a hyperplane, and c_i is non-zero.

Griffiths transversality places the derivative in $H^{2,0}(S) \oplus H^{1,1}_{\mathrm{prim}}(S)$; Shepherd–Barron's formula further shows that, for each Schiffer direction, the $(1,1)$-components of the derivatives of all holomorphic two-forms lie in the same line $\mathbb{C}\eta_i$. The kernel description identifies c_i with evaluation at p_i, up to a non-zero scalar, giving the following formula.

Lemma 3.4. *For $f \in H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(n-2))$ one has*

$$c_i(f\,\omega_0) = c_i(\omega_0) f(p_i) \qquad c_i(\omega_0) \neq 0$$

In particular, $c_i(\omega_j) = c_i(\omega_0) z_i^j$ for $0 \leq j \leq h-1$.

Proof. Under the polynomial identification above, write $f = f(p_i) + (f - f(p_i))$. Since $f - f(p_i)$ vanishes at p_i, the form $(f - f(p_i))\omega_0$ belongs to $\ker(c_i)$. Linearity therefore gives $c_i(f\,\omega_0) = f(p_i) c_i(\omega_0)$. Since ω_0 does not vanish along E_{p_i}, it does not belong to $\ker(c_i)$, so $c_i(\omega_0) \neq 0$. Taking $f(z) = z^j$ gives the final formula. ∎

The evaluation formula lets us test whether the h period equations for a fixed class have independent differentials. The following criterion does this by selecting h ramification points.

Theorem 3.5. *Let γ be a locally flat real class which is of type $(1,1)$ at $[S] \in \mathcal{W}_n^{\mathrm{sb}}$. If there are distinct indices $i_1, \ldots, i_h$ such that*

$$\gamma \cdot \eta_{i_k} \neq 0 \qquad (1 \leq k \leq h)$$

then, on the marked local deformation space, the fixed-class type-$(1,1)$ locus defined by γ is smooth of codimension $h = n-1$ at $[S]$.

Proof. Extend $\omega_0, \ldots, \omega_{h-1}$ to a holomorphic frame of F^2 near $[S]$ and set $f_j = \gamma \cdot \omega_j$ for $0 \leq j < h$. For a real flat class, vanishing of these pairings is equivalent to being of type $(1,1)$; the conjugate equations follow automatically. Flatness of γ and compatibility of the Gauss–Manin connection with the intersection pairing give $\partial f_j / \partial t_i = \gamma \cdot \nabla_{\partial/\partial t_i} \omega_j$. At $[S]$, the $H^{2,0}(S)$-component of the derivative pairs trivially with γ, since γ has type $(1,1)$. Shepherd–Barron's formula and Lemma 3.4 therefore give

$$\left.\frac{\partial f_j}{\partial t_{i_k}}\right|_{[S]} = c_{i_k}(\omega_0) z_{i_k}^j (\gamma \cdot \eta_{i_k})$$

If J_γ denotes this $h \times h$ minor, then

$$\det J_\gamma = \left(\prod_{k=1}^{h} c_{i_k}(\omega_0)(\gamma \cdot \eta_{i_k}) \right) \prod_{1 \leq a < b \leq h} (z_{i_b} - z_{i_a})$$

The first product is non-zero by Lemma 3.4 and the hypothesis on γ. The second is non-zero because the ramification points are distinct. Hence the Jacobian of $(f_0, \dots, f_{h-1})$ has rank h at $[S]$. These equations define the fixed-class type-$(1,1)$ locus, so the holomorphic implicit function theorem makes this locus smooth of codimension h at $[S]$. ∎

The same period calculation shows that every non-zero fixed integral class in Λ_n has a proper Hodge locus on a marked local chart. Together with the algebraicity of Hodge loci, this gives the following well-known statement. For $n \geq 3$, it follows from Cox's result that a very general surface has Picard number two [Cox90, Prop. 3.2]; for $n = 2$, it is classical. We include a proof using Schiffer deformation coordinates.

Proposition 3.6. *For every $n \geq 2$, a very general member of $\mathcal{W}_{\min,n}$ has Mordell–Weil rank zero.*

Proof. At any point of $\mathcal{W}_n^{\mathrm{sb}}$, every non-zero integral Hodge class in Λ_n pairs non-trivially with some Schiffer image class. The rank-one formula therefore gives a deformation direction in which one of its periods has non-zero derivative. Consequently, every non-zero fixed-class Hodge locus on a marked local chart is a proper closed analytic subset. Since the lattice is countable, Baire's theorem shows that their union has empty analytic interior.

Apply the algebraicity theorem of Cattani–Deligne–Kaplan [CDK95, Thm. 1.1 and Cor. 1.3] to the Tate twist of the primitive variation on a smooth algebraic atlas of finite type for $\mathcal{W}_n^{\mathrm{sb}}$, so that primitive $(1,1)$-classes become classes of type $(0,0)$. The locus carrying a non-zero integral Hodge class is a countable union of closed algebraic subsets of this atlas. Their images in $\mathcal{W}_n^{\mathrm{sb}}$ are constructible. None is dense, since a dense constructible image would contain a non-empty open subset, contradicting the local empty-interior assertion.

Their closures in $\mathcal{W}_{\min,n}$, together with the proper closed complement of $\mathcal{W}_n^{\mathrm{sb}}$, form a countable union of proper closed substacks. Outside this union there are no non-zero integral Hodge classes in Λ_n. Since all singular fibres are irreducible, the Shioda–Tate formula gives Mordell–Weil rank zero. ∎

For the height bounds, we need a stronger local avoidance statement: on a relatively compact marked chart, only finitely many non-zero integral classes with bounded $-\gamma^2/2$ can become Hodge.

Lemma 3.7. *Let B_0 be a marked local deformation chart in $\mathcal{W}_n^{\mathrm{sb}}$, and let $B \subset B_0$ be a connected open subset whose closure is compact and contained in B_0. For every $H > 0$, the locus*

$$B_{\leq H} = \left\{ b \in B : \text{there exists } 0 \neq \gamma \in \Lambda_n \cap H^{1,1}(S_b) \text{ with } -\frac{1}{2}\gamma^2 \leq H \right\}$$

is a proper closed analytic subset of B. Its complement $B_{>H}$ is therefore open and dense.

Proof. Fix a Euclidean norm on $\Lambda_{n,\mathbb{R}}$. The primitive real $(1,1)$-spaces vary continuously over $\overline{B}$, and the intersection pairing is negative definite on each of them.

Compactness therefore gives a constant $c > 0$ such that

$$-\gamma^2 \geq c\|\gamma\|^2 \qquad \left(b \in \overline{B},\ \gamma \in H^{1,1}_{\mathrm{prim}}(S_b, \mathbb{R})\right)$$

Consequently, every non-zero integral Hodge class with $-\gamma^2/2 \leq H$ belongs to the finite set

$$A_H = \left\{\gamma \in \Lambda_n : 0 < -\frac{1}{2}\gamma^2 \leq H,\ \|\gamma\|^2 \leq \frac{2H}{c}\right\}$$

Thus $B_{\leq H}$ is the union of the fixed-class Hodge loci of the classes in A_H. Each is closed analytic and proper by the period calculation in the proof of Proposition 3.6. Their finite union is proper because B is connected. ∎

3.3. **Simultaneous period equations.** For r classes, we need rh period equations with independent differentials. We first construct a complex tuple for which the Jacobian splits into r Vandermonde blocks, then use polynomial non-vanishing to obtain a real tuple.

Lemma 3.8. *Let $r \geq 1$, put $M = rh$, and suppose $M \leq N$. Put $W := H^{1,1}_{\mathrm{prim}}(S, \mathbb{R})$. There exist $\mathbb{R}$-linearly independent vectors $v_1, \ldots, v_r \in W$ with $v_a^2 = -1$ for $1 \leq a \leq r$, and a subset $I \subset \{1, \ldots, N\}$ of cardinality M, such that the $M \times M$ matrix*

$$J_{(a,j),i} = c_i(\omega_0) z_i^j (v_a \cdot \eta_i) \qquad 1 \leq a \leq r, \quad 0 \leq j \leq h-1, \quad i \in I$$

is non-singular.

Proof. Choose any M indices and partition them into disjoint blocks $I = C_1 \sqcup \cdots \sqcup C_r$ with $|C_a| = h$. Over $\mathbb{C}$, set

$$v_a^\circ = \sum_{i \in C_a} \frac{\eta_i}{\eta_i^2}$$

The denominators are non-zero because the η_i form an orthogonal basis of a non-degenerate quadratic space. Orthogonality gives

$$v_a^\circ \cdot \eta_i = \begin{cases} 1, & i \in C_a \\ 0, & i \notin C_a \end{cases}$$

After ordering rows and columns by the blocks, the resulting matrix is block diagonal. Its a-th block has determinant

$$\left(\prod_{i \in C_a} c_i(\omega_0)\right) \prod_{\substack{i,k \in C_a \\ i<k}} (z_k - z_i)$$

which is non-zero.

Thus the determinant is a non-zero polynomial on $(W \otimes_{\mathbb{R}} \mathbb{C})^r$. A complex polynomial which vanishes on the real form W^r vanishes identically. Hence we may choose a real tuple on which the determinant is non-zero. The chosen vectors are linearly independent: any relation $\sum_a \lambda_a v_a = 0$ would give the same relation among the rows indexed by $(a, 0)$, contradicting determinant non-vanishing.

Finally, negative definiteness allows us to replace each v_a by $v_a/\sqrt{-v_a^2}$. This multiplies its h corresponding rows by the same non-zero scalar, so the determinant remains non-zero. ∎

To pass from the real tuple to integral classes, we approximate its directions by indivisible lattice vectors of increasing norm. These approximations may leave the original Hodge subspace; we will subsequently deform the surface to make the chosen integral classes Hodge.

Lemma 3.9. *Let $L \subset E$ be a full lattice, meaning a discrete subgroup spanning the finite-dimensional real vector space E. Assume $\dim E \geq 2$ and equip E with a Euclidean norm. Let $v_1, \ldots, v_r \in E$ be linearly independent unit vectors. There exist indivisible vectors $\gamma_{a,m} \in L$, for $1 \leq a \leq r$, such that*

$$\|\gamma_{a,m}\| \longrightarrow \infty \qquad \frac{\gamma_{a,m}}{\|\gamma_{a,m}\|} \longrightarrow v_a$$

and, for all sufficiently large m, the vectors $\gamma_{1,m}, \ldots, \gamma_{r,m}$ are linearly independent. If a proper real subspace $E_0 \subsetneq E$ contains all the v_a, the vectors $\gamma_{a,m}$ may moreover be chosen outside E_0.

Proof. After choosing a basis of L, the directions of indivisible lattice vectors are precisely the rational directions, which are dense in the unit sphere of E. For each m, only finitely many indivisible vectors have norm at most m; deleting their finitely many directions does not destroy density.

If $E_0 \subsetneq E$ is a proper real subspace, rational directions outside E_0 are still dense: every open subset of the unit sphere contains a smaller open subset disjoint from E_0, and rational directions are dense there. For each a and m, we may therefore choose an indivisible vector $\gamma_{a,m}$, outside E_0 when specified, such that

$$\|\gamma_{a,m}\| > m \qquad \left\| \frac{\gamma_{a,m}}{\|\gamma_{a,m}\|} - v_a \right\| < \frac{1}{m}$$

These inequalities give the two asserted limits. Since linear independence is an open condition on tuples of vectors, convergence to the independent tuple $(v_1, \ldots, v_r)$ gives independence for all sufficiently large m. ∎

3.4. **Local simultaneous realization.** The implicit function theorem lets us compensate for a small change in the cohomology classes by a small deformation of the surface. We apply this to indivisible lattice vectors whose normalized directions approach the real tuple constructed above, while their norms tend to infinity.

Theorem 3.10. *Let $n \geq 2$, let $[S] \in \mathcal{W}_n^{\mathrm{sb}}$, and let $r \geq 1$ satisfy $r(n-1) \leq 10n-2$. Then there are points $b_m \to 0$ in the marked local deformation chart B chosen above, representing pairwise non-isomorphic Jacobian elliptic surfaces,*

$$[S_m] \in \mathcal{W}_n^{\mathrm{sb}} \qquad [S_m] \longrightarrow [S]$$

and, after identifying the primitive cohomology lattices by flat transport, indivisible classes $\gamma_{1,m}, \ldots, \gamma_{r,m} \in \Lambda_n$ with the following properties:

1. *every $\gamma_{a,m}$ is of type $(1,1)$ on S_m;*
2. *the classes $\gamma_{1,m}, \ldots, \gamma_{r,m}$ span a negative-definite subspace and are linearly independent;*
3. *on the marked local deformation space, the $r(n-1)$ period equations defining their simultaneous fixed-class Hodge locus have linearly independent differentials at $[S_m]$; in particular, this locus is smooth of complex codimension $r(n-1)$ there;*

(4) *for every a,*

$$d_{a,m} := -\frac{\gamma_{a,m}^2}{2} \in \mathbb{Z}_{>0} \qquad d_{a,m} \longrightarrow \infty$$

Proof. Put $M = rh$. Work on the simply-connected chart B above. Flat transport identifies the real primitive cohomology of its fibres with $\Lambda_{n,\mathbb{R}}$. At the central surface, Hodge decomposition gives an orthogonal splitting

$$\Lambda_{n,\mathbb{R}} = H_+ \oplus W$$

where $H_+ = \left(H^{2,0}(S) \oplus H^{0,2}(S)\right) \cap \Lambda_{n,\mathbb{R}}$ and $W = H^{1,1}_{\mathrm{prim}}(S,\mathbb{R})$. The intersection form is positive definite on H_+ and negative definite on W. We therefore use the positive-definite norm determined by

$$\|v+w\|^2 = v^2 - w^2 \qquad v \in H_+, \quad w \in W$$

In particular, the vectors supplied by Lemma 3.8 have norm one.

Choose $v_1,\dots,v_r \in W$ and $I \subset \{1,\dots,N\}$ as in Lemma 3.8. Let $s = (t_i)_{i\in I}$ be the corresponding M coordinates, and let $B_I \subset B$ be the slice on which all other coordinates are zero. Restrict B_I to a sufficiently small coordinate polydisc $\mathbb{D}^M \subset B$ with coordinates s, where $\mathbb{D} \subset \mathbb{C}$ is an open disc centred at zero. Choose a holomorphic frame $\omega_0(t),\dots,\omega_{h-1}(t)$ of F^2 over B whose value at the origin is the monomial basis above. With the notation $\omega_j(s) = \omega_j(t)|_{B_I}$, define the restriction of the Hodge-equation map to this slice by

$$\Psi\colon (\Lambda_{n,\mathbb{C}})^r \times \mathbb{D}^M \longrightarrow \mathbb{C}^{rh}$$
$$\Psi_{a,j}(\boldsymbol{u},s) = u_a \cdot \omega_j(s) \qquad 1 \le a \le r, \quad 0 \le j \le h-1$$

Because $v_a \in W$, one has $\Psi(\boldsymbol{v},0) = 0$. Moreover, the same calculation as in the proof of Theorem 3.5, using Lemma 3.4, gives

$$\left.\frac{\partial \Psi_{a,j}}{\partial t_i}\right|_{(\boldsymbol{v},0)} = c_i(\omega_0) z_i^j (v_a \cdot \eta_i)$$

so the derivative with respect to s is non-singular by Lemma 3.8.

The holomorphic implicit function theorem therefore gives a neighbourhood $\mathcal{V}$ of $\boldsymbol{v} = (v_1,\dots,v_r)$ in $(\Lambda_{n,\mathbb{C}})^r$ and a holomorphic map $\tau\colon \mathcal{V} \to \mathbb{D}^M$ with $\tau(\boldsymbol{v}) = 0$ such that

$$\Psi(\boldsymbol{u}, \tau(\boldsymbol{u})) = 0 \tag{3}$$

for every $\boldsymbol{u} \in \mathcal{V}$. Shrinking $\mathcal{V}$ if necessary, the derivative with respect to s remains non-singular along the graph of τ.

The period equations are homogeneous in each class, so normalization does not change their zero locus. Since $h > 0$, the positive subspace H_+ is non-zero and $W \subset \Lambda_{n,\mathbb{R}}$ is proper. We may therefore apply Lemma 3.9 to the lattice $\Lambda_n \subset \Lambda_{n,\mathbb{R}}$ with $E_0 = W$. Choose indivisible vectors $\gamma_{a,m} \notin W$ such that

$$u_{a,m} := \frac{\gamma_{a,m}}{\|\gamma_{a,m}\|} \longrightarrow v_a \qquad \|\gamma_{a,m}\| \longrightarrow \infty$$

For large m, the tuple $\boldsymbol{u}_m$ lies in $\mathcal{V}$. Let $b_m \in B$ have selected coordinates $\tau(\boldsymbol{u}_m)$ and all remaining coordinates zero, and let S_m be the corresponding surface. Then

$b_m \to 0$, and (3) yields

$$\gamma_{a,m} \cdot \omega = 0 \qquad \text{for every } \omega \in F^2(S_m) \quad \text{and every } 1 \le a \le r$$

The classes $\gamma_{a,m}$ are real, so they are also orthogonal to $\overline{F^2(S_m)}$ and hence are of type $(1,1)$.

For the fixed tuple $(\gamma_{1,m}, \dots, \gamma_{r,m})$, the simultaneous Hodge locus in the full chart B is defined by the rh equations

$$\gamma_{a,m} \cdot \omega_j(t) = 0 \qquad 1 \le a \le r, \quad 0 \le j \le h-1$$

Their Jacobian contains the selected $M \times M$ minor from the implicit function argument. Replacing $u_{a,m}$ by $\gamma_{a,m}$ multiplies each of the h rows belonging to a by $\|\gamma_{a,m}\|$, so this minor remains non-singular. The full Jacobian therefore has rank $M = rh$, and the simultaneous fixed-class Hodge locus is smooth of codimension M at $[S_m]$.

Each $\gamma_{a,m}$ is a non-zero class in $H^{1,1}_{\mathrm{prim}}(S_m, \mathbb{R})$, where the intersection pairing is negative definite. Thus their span is negative definite, and each $d_{a,m}$ is positive. By Lemma 3.9, the classes are linearly independent for all sufficiently large m. Each $d_{a,m}$ is an integer because Λ_n is even. Since $v_a^2 = -1$, one has

$$\frac{d_{a,m}}{\|\gamma_{a,m}\|^2} = -\frac{\gamma_{a,m}^2}{2\|\gamma_{a,m}\|^2} \longrightarrow \frac{1}{2}$$

Therefore $d_{a,m} \to \infty$.

Finally, $\gamma_{a,m} \notin W$ implies that it is not of type $(1,1)$ at the central surface, so $\tau(\boldsymbol{u}_m) \neq 0$. These non-zero points tend to zero; after passing to a subsequence they are pairwise distinct. Since G is finite, a further subsequence lies in pairwise distinct G-orbits. By the choice of the uniformizing chart, their images in the coarse analytic moduli space are pairwise distinct; equivalently, the corresponding Jacobian elliptic surfaces are pairwise non-isomorphic. All the resulting points lie in $\mathcal{W}_n^{\mathrm{sb}}$ by construction. After discarding finitely many initial terms and relabelling the final subsequence, properties (1)–(4) hold for every m. ∎

3.5. **Mordell–Weil rank, section heights, and analytic density.** Since every fibre is irreducible, the Shioda–Tate description removes only the fibre and zero-section classes. The primitive integral Hodge classes constructed above therefore give Mordell–Weil sections, and their intersection pairing determines the canonical heights. By avoiding bounded-height Hodge classes before applying local realization, we make the least non-zero section height tend to infinity. We then use density of the simply-branched locus to obtain statements throughout the minimal elliptic locus.

Theorem 3.11. *Let n, r be integers satisfying $n \ge 2$ and*

$$1 \le r \le \left\lfloor \frac{10n-2}{n-1} \right\rfloor$$

For every $[S] \in \mathcal{W}_n^{\mathrm{sb}}$ there are pairwise non-isomorphic Jacobian elliptic surfaces

$$[S_m] \in \mathcal{W}_n^{\mathrm{sb}} \qquad [S_m] \longrightarrow [S]$$

and linearly independent sections $P_{1,m},\dots,P_{r,m}\in \mathrm{MW}(S_m/\mathbb{P}^1)$ *such that*

$$\min_{0\neq P\in \mathrm{MW}(S_m/\mathbb{P}^1)} \widehat{h}(P)\longrightarrow\infty$$

In particular, $\mathrm{rk}\,\mathrm{MW}(S_m/\mathbb{P}^1)\geq r$. *The corresponding Shioda classes are indivisible in* Λ_n*, and the canonical heights of non-zero sections are positive integers. On the marked local deformation space, the* $r(n-1)$ *period equations defining the simultaneous fixed-class Hodge locus of the chosen classes have linearly independent differentials at* $[S_m]$*; hence this locus is smooth of complex codimension* $r(n-1)$ *there.*

Proof. Choose a marked local deformation chart B centred at a point b_0 representing $[S]$, relatively compact in a larger marked chart, and a decreasing neighbourhood basis U_m of b_0. By Lemma 3.7, each $U_m\cap B_{>m}$ is non-empty and open. Choose a point in this intersection and apply Theorem 3.10 on a sufficiently small marked deformation chart contained in $U_m\cap B_{>m}$. Its sequence contains infinitely many pairwise non-isomorphic surfaces, and all sufficiently late terms have representatives in $U_m\cap B_{>m}$.

Choose one such representative b_m, with $S_m=S_{b_m}$ not isomorphic to any previously chosen surface, together with its r independent indivisible integral Hodge classes $\gamma_{1,m},\dots,\gamma_{r,m}$. Use the marking on B to regard these classes as elements of Λ_n. Then $b_m\to b_0$, and the simultaneous period differentials are independent at b_m. Put $d_{a,m}=-\gamma_{a,m}^2/2$. Since $b_m\in B_{>m}$, we have $d_{a,m}>m$ for every a.

Since S_m has singular fibres, its Néron–Severi group is torsion-free [Shi90, Thm. 1.2]. We therefore identify $\mathrm{NS}(S_m)$ with its image in $H^2(S_m,\mathbb{Z})/\mathrm{tors}$. The Lefschetz $(1,1)$-theorem gives $\gamma_{a,m}\in\mathrm{NS}(S_m)$ for every a [CMSP17, §3.4, pp. 119–120].

All fibres of S_m are irreducible, so $\mathrm{Triv}(S_m)=\langle F,O\rangle$. Since this lattice is unimodular and F and O are algebraic, the integral orthogonal splitting from Section 2 restricts to

$$\mathrm{NS}(S_m)=\mathrm{Triv}(S_m)\oplus\big(\mathrm{NS}(S_m)\cap\Lambda_n\big)$$

The Shioda–Tate description [Shi90, Thm. 1.3] therefore gives

$$\mathrm{MW}(S_m/\mathbb{P}^1)\simeq \mathrm{NS}(S_m)/\mathrm{Triv}(S_m)\simeq \mathrm{NS}(S_m)\cap\Lambda_n$$

In particular, $\mathrm{MW}(S_m/\mathbb{P}^1)$ is torsion-free. Under this identification, the map from sections to algebraic classes in Λ_n is Shioda's orthogonal projection σ_{Sh} [Shi90, Lem. 8.1]. Consequently, each $\gamma_{a,m}$ determines a unique section $P_{a,m}$ with $\sigma_{\mathrm{Sh}}(P_{a,m})=\gamma_{a,m}$. Since this correspondence is an isomorphism of abelian groups, the linear independence of the classes gives the linear independence of the sections.

Shioda's height pairing [Shi90, Thm. 8.4], with our half-normalization, gives

$$\widehat{h}(P)=-\frac{1}{2}\,\sigma_{\mathrm{Sh}}(P)^2$$

In particular, $\widehat{h}(P_{a,m})=d_{a,m}>m$, so the heights of the chosen sections tend to infinity.

More generally, every non-zero section $P\in\mathrm{MW}(S_m/\mathbb{P}^1)$ has a non-zero integral Shioda class of type $(1,1)$. Since $b_m\in B_{>m}$, every such section satisfies $\widehat{h}(P)>m$. The lattice Λ_n is even and its real primitive $(1,1)$-subspace is negative definite, so these heights are positive integers. The Mordell–Weil group has positive rank,

hence the set of non-zero section heights is non-empty and has a minimum. Therefore

$$\min_{0\neq P\in \mathrm{MW}(S_m/\mathbb{P}^1)} \widehat{h}(P) > m$$

which proves the asserted divergence.

The chosen classes were selected from applications of Theorem 3.10. Their indivisibility and the independence of their simultaneous period differentials are therefore preserved. The holomorphic implicit function theorem gives the stated smoothness and codimension of their simultaneous fixed-class Hodge locus. ■

Corollary 3.12. *Let $n \geq 2$. Every analytic neighbourhood of every point of $\mathcal{W}_{\min,n}$ contains a sequence of pairwise non-isomorphic Jacobian elliptic surfaces $[S_m] \in \mathcal{W}_n^{\mathrm{sb}}$ carrying sections $P_m \in \mathrm{MW}(S_m/\mathbb{P}^1)$ with indivisible Shioda classes such that the canonical heights $\widehat{h}(P_m)$ form a strictly increasing sequence of positive integers.*

Proof. Let U be an analytic neighbourhood of any point of $\mathcal{W}_{\min,n}$. By Lemma 3.3, choose $[S] \in U \cap \mathcal{W}_n^{\mathrm{sb}}$. Apply Theorem 3.11 at $[S]$ with $r = 1$. The resulting pairwise non-isomorphic surfaces converge to $[S]$, so all sufficiently late members lie in U. By Theorem 3.11, the chosen sections have indivisible Shioda classes, and their canonical heights are positive integers tending to infinity. Passing to a subsequence makes these heights strictly increasing. ■

Corollary 3.13. *Let $n \geq 2$ and $r \geq 1$ satisfy $r(n-1) \leq 10n-2$. The locus*

$$\left\{[S] \in \mathcal{W}_n^{\mathrm{sb}} : \mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1) \geq r\right\}$$

is analytically dense in $\mathcal{W}_{\min,n}$.

More precisely, for every non-empty analytic open subset $U \subset \mathcal{W}_{\min,n}$ and every $H > 0$, there are infinitely many pairwise non-isomorphic surfaces $[S'] \in U \cap \mathcal{W}_n^{\mathrm{sb}}$ carrying r independent sections with indivisible Shioda classes and satisfying $\widehat{h}(P) > H$ for every $0 \neq P \in \mathrm{MW}(S'/\mathbb{P}^1)$.

The $r(n-1)$ simultaneous fixed-class period equations are transverse at $[S']$. On a marked local deformation chart $B \to \mathcal{W}_n^{\mathrm{sb}}$ at $[S']$, their Hodge germ has a smooth representative $Z \subset B$ of dimension $10n-2-r(n-1)$, with image in $U \cap \mathcal{W}_n^{\mathrm{sb}}$, such that

$$\widehat{h}(P) > H \qquad \text{for every } b \in Z \text{ and every } 0 \neq P \in \mathrm{MW}(S_b/\mathbb{P}^1)$$

Proof. By Lemma 3.3, choose $[S] \in U \cap \mathcal{W}_n^{\mathrm{sb}}$. Theorem 3.11 gives pairwise non-isomorphic surfaces $[S_m] \to [S]$ carrying the required independent sections and with least non-zero section height tending to infinity. All sufficiently late terms lie in U and satisfy the prescribed bound. Indivisibility and transversality follow from the same theorem.

At each selected surface, the integral Shioda identification shows that there is no non-zero integral primitive-cohomology Hodge class with $-\gamma^2/2 \leq H$. Lemma 3.7 therefore gives an open neighbourhood with the same property. Shrinking the simultaneous Hodge germ into this neighbourhood and into the inverse image of U gives the required representative. ■

For $r = 1$, the density and arbitrarily large lower bounds for all non-zero section heights recover Cox's theorem [Cox90, Thm. 4.4]; on the all-I_1 locus, his height $P\cdot O$

and our normalization satisfy $\widehat{h}(P) = n + P \cdot O$. The proof above supplies explicit transverse period minors and smooth Hodge locus representatives on which all non-zero section heights satisfy a prescribed uniform lower bound. Subsection 3.8 identifies the corresponding section-moduli germs.

3.6. **Exact Mordell–Weil ranks and first-order criteria.** We prove exact-rank density together with uniform lower bounds for the heights of all non-zero sections. We state the conclusion first; its proof is assembled below, with the rank-10 case supplied by the next subsection.

Theorem 3.14. *Let $n \geq 2$ and $r \geq 1$ be integers, with $r \leq 18$ if $n = 2$ and $r \leq 10$ if $n \geq 3$. Then the locus*

$$\left\{[S] \in \mathcal{W}_n^{\mathrm{sb}} : \mathrm{rk}\,\mathrm{MW}(S/\mathbb{P}^1) = r\right\}$$

is analytically dense in $\mathcal{W}_{\min,n}$.

More precisely, for every non-empty analytic open subset $U \subset \mathcal{W}_{\min,n}$ and every $H > 0$, there are a marked deformation chart $B \to \mathcal{W}_n^{\mathrm{sb}}$ parametrizing surfaces S_b and a smooth connected local simultaneous fixed-class Hodge locus $Z \subset B$ of dimension $10n-2-r(n-1)$, defined by r independent indivisible integral classes, with image in $U \cap \mathcal{W}_n^{\mathrm{sb}}$, such that the Mordell–Weil rank is exactly r outside a countable union of proper closed analytic subsets of Z and

$$\widehat{h}(P) > H \qquad \text{for every } b \in Z \text{ and every } 0 \neq P \in \mathrm{MW}(S_b/\mathbb{P}^1)$$

The imposed classes already give a lower bound on Mordell–Weil rank. To obtain exact rank, we must prevent any integral class outside their rational span from remaining Hodge throughout the constructed family. On a connected smooth simultaneous Hodge locus, each additional integral class either remains Hodge everywhere or has a proper closed analytic Hodge locus. If the first possibility is excluded for every such class, countability of the lattice and Baire's theorem give exact rank outside a countable union of proper closed analytic subsets.

We will exclude persistent additional classes by differentiating their period equations along the imposed Hodge locus. A non-zero restricted derivative shows that the corresponding class cannot remain Hodge identically. This first-order method suffices for $1 \leq r \leq 9$ when $n \geq 3$ and for $1 \leq r \leq 18$ when $n = 2$. We prove the remaining case $r = 10$ when $n \geq 3$ by the second-order argument of Subsection 3.7.

The infinitesimal principle is classical; compare conditions (1) and (3) and the proof of [Voi07, Thm. 3.1]. The density step is a local form of Green's criterion, recalled in [Voi07, Prop. 3.3]. Shepherd–Barron's frame makes the required derivative conditions explicit.

Keep $n \geq 2$, $h = n-1$, and $N = 10n-2$. Work on a sufficiently small full marked deformation chart B in $\mathcal{W}_n^{\mathrm{sb}}$, centred at 0, and write S_b for the surface over $b \in B$. Use the Schiffer tangent basis, the distinct finite ramification coordinates $z_1, \ldots, z_N$, and the basis $\eta_1, \ldots, \eta_N$ of $H^{1,1}_{\mathrm{prim}}(S_0)$ constructed above. Extend $\omega_0, \ldots, \omega_{h-1}$ to a holomorphic frame of F^2 over B. For a flat real class $\gamma \in \Lambda_{n,\mathbb{R}}$, define

$$\Phi_\gamma(b) = \big(\gamma \cdot \omega_0(b), \ldots, \gamma \cdot \omega_{h-1}(b)\big) \qquad Z_\gamma = \Phi_\gamma^{-1}(0) \tag{4}$$

Since γ is real, Z_γ is precisely its type-$(1,1)$ locus.

If γ is Hodge at 0, Lemma 3.4 identifies its period Jacobian in the Schiffer tangent basis as

$$\begin{aligned} \mathrm{d}\Phi_\gamma(0) = A_\gamma = V\,\mathrm{diag}\big(c_i(\omega_0)(\gamma\cdot\eta_i)\big)_{i=1}^{N} \\ V = (z_i^j)_{\substack{0\le j<h \\ 1\le i\le N}} \end{aligned} \tag{5}$$

We keep the non-zero factors $c_i(\omega_0)$ from the preceding construction.

We begin with one imposed class α. If α is Hodge at 0 and pairs non-trivially with every η_i, then (5) shows that A_α is surjective and $T_0Z_\alpha = \ker A_\alpha$. For an additional integral class β that is Hodge at 0, it suffices to find a tangent vector $v \in \ker A_\alpha$ with $A_\beta(v) \neq 0$. This preserves α to first order while changing the Hodge type of β, so β cannot remain Hodge throughout Z_α.

We will prove that $\ker A_\alpha \subseteq \ker A_\beta$ can hold only when β is proportional to α. In (5), put $a_i = c_i(\omega_0)(\alpha\cdot\eta_i)$ and $b_i = c_i(\omega_0)(\beta\cdot\eta_i)$. The following lemma shows that the kernel inclusion forces $b = ca$, which translates back into proportionality of the classes.

Lemma 3.15. *Let $z_1,\dots,z_N$ be distinct complex numbers, let $h \ge 1$, and assume $N \ge h$ and $N > 2h-2$. Set $V = (z_i^j)_{0\le j<h,\,1\le i\le N}$. For $a \in (\mathbb{C}^*)^N$ and $b \in \mathbb{C}^N$, put $A = V\,\mathrm{diag}(a)$ and $C = V\,\mathrm{diag}(b)$. If $\ker A \subseteq \ker C$, then $b = ca$ for some $c \in \mathbb{C}$.*

Proof. The first h columns of V have non-zero determinant $\prod_{1\le i<k\le h}(z_k - z_i)$, so A has row rank h. Every row of C annihilates $\ker A$, hence belongs to the row space of A. A vector in that row space has entries $a_i p(z_i)$ for a polynomial p of degree at most $h-1$. Applying this to the first row of C gives

$$b_i = a_i p(z_i) \qquad (1 \le i \le N) \tag{6}$$

For $h = 1$, this proves the assertion.

For $h \ge 2$, apply the same argument to the last row of C. There is a polynomial q of degree at most $h-1$ satisfying

$$z_i^{h-1} b_i = a_i q(z_i) \qquad (1 \le i \le N)$$

Dividing by $a_i \neq 0$ and using (6), we find that $z^{h-1}p(z) - q(z)$ vanishes at all N points z_i. Its degree is at most $2h-2 < N$, so it vanishes identically. If $p \neq 0$, the identity $z^{h-1}p = q$ gives $h - 1 + \deg p = \deg q \le h-1$, so p is constant. If $p = 0$, then $b = 0$. In either case $b = ca$. ∎

We now apply this lemma to period derivatives. The non-vanishing condition on α will ensure that every independent additional integral class is detected by a tangent direction along its Hodge locus.

Proposition 3.16. *Let $\alpha \in \Lambda_n$ be non-zero, indivisible in Λ_n, and of type $(1,1)$ on S_0. Assume*

$$\alpha\cdot\eta_i \neq 0 \qquad (1 \le i \le N) \tag{7}$$

Then Z_α is smooth of complex codimension $h = n-1$ at 0. For every $\beta \in \Lambda_n \setminus \mathbb{Q}\alpha$ that is Hodge at 0, one has

$$\mathrm{d}\Phi_\beta(0)\big|_{T_0Z_\alpha} \neq 0 \tag{8}$$

After replacing Z_α by a sufficiently small connected representative of its germ, there is a countable union $\mathcal{E}$ of proper closed analytic subsets of Z_α such that

$$\Lambda_n \cap H^{1,1}(S_b) = \mathbb{Z}\alpha \qquad \operatorname{rk}\mathrm{MW}(S_b/\mathbb{P}^1) = 1 \qquad (b \in Z_\alpha \smallsetminus \mathcal{E}) \tag{9}$$

These points are dense in Z_α and have full measure for every measure absolutely continuous with respect to intrinsic smooth volume on Z_α.

Proof. Condition (7) allows us to apply Theorem 3.5 to α. Its Jacobian calculation shows that $\mathrm{d}\Phi_\alpha(0)$ is surjective, so Z_α is smooth of codimension h and

$$T_0 Z_\alpha = \ker A_\alpha \tag{10}$$

Shrink the chart so that Z_α is a connected complex manifold.

Let $\beta \in \Lambda_n \smallsetminus \mathbb{Q}\alpha$ be Hodge at 0. If (8) failed, then $\ker A_\alpha \subseteq \ker A_\beta$. Apply Lemma 3.15 with $a_i = c_i(\omega_0)(\alpha \cdot \eta_i)$ and $b_i = c_i(\omega_0)(\beta \cdot \eta_i)$. The numerical hypotheses hold since $N \geq h$ and $N-(2h-2) = 8n+2 > 0$. The lemma gives $b_i = c a_i$ for some $c \in \mathbb{C}$. Cancelling $c_i(\omega_0) \neq 0$, we obtain

$$(\beta - c\alpha) \cdot \eta_i = 0 \qquad (1 \leq i \leq N)$$

Both α and β lie in $H^{1,1}_{\mathrm{prim}}(S_0)$. The intersection pairing on this space is non-degenerate: its real form is negative definite by the Hodge index theorem. Since the η_i form a basis, the displayed equalities force $\beta = c\alpha$ in $\Lambda_{n,\mathbb{C}}$. Comparing an integral lattice coordinate in which α is non-zero gives $c \in \mathbb{Q}$, a contradiction. This proves (8).

For each $\beta \in \Lambda_n \smallsetminus \mathbb{Q}\alpha$, the subset $Z_\alpha \cap Z_\beta$ is closed analytic in Z_α. If β is not Hodge at 0, this subset does not contain 0 and is therefore proper. If β is Hodge at 0, equation (8) shows that Φ_β cannot vanish identically on Z_α, so the subset is again proper. Thus

$$\mathcal{E} = \bigcup_{\beta \in \Lambda_n \smallsetminus \mathbb{Q}\alpha} (Z_\alpha \cap Z_\beta)$$

is a countable union of proper closed analytic subsets. Each is nowhere dense and has measure zero with respect to every smooth positive density on Z_α. Baire's theorem gives density of the complement, and countable additivity gives the measure assertion.

For $b \notin \mathcal{E}$, every integral Hodge class in Λ_n belongs to $\mathbb{Q}\alpha$. Since α is indivisible, $\Lambda_n \cap \mathbb{Q}\alpha = \mathbb{Z}\alpha$. The integral Shioda identification established in the proof of Theorem 3.11, valid for every surface in $\mathcal{W}_n^{\mathrm{sb}}$, therefore gives $\mathrm{MW}(S_b/\mathbb{P}^1) \simeq \Lambda_n \cap H^{1,1}(S_b) = \mathbb{Z}\alpha$. This proves (9). ∎

To obtain density, we must produce surfaces carrying a class that satisfies the preceding proposition. We first choose a real Hodge class with all the required pairings non-zero. We then approximate it by rational classes in the full primitive real cohomology and use the implicit function theorem to make those classes Hodge on nearby surfaces.

Corollary 3.17. *For every $n \geq 2$, the locus*

$$\left\{[S] \in \mathcal{W}_n^{\mathrm{sb}} : \operatorname{rk}\mathrm{MW}(S/\mathbb{P}^1) = 1\right\}$$

is dense in $\mathcal{W}_{\min,n}$ for the complex analytic topology.

Proof. Let $U \subset \mathcal{W}_{\min,n}$ be a non-empty analytic open subset. By Lemma 3.3, choose $[S_0] \in U \cap \mathcal{W}_n^{\mathrm{sb}}$ and a full marked deformation chart B centred at 0 whose image lies in $U \cap \mathcal{W}_n^{\mathrm{sb}}$.

Each functional $\lambda \mapsto \lambda \cdot \eta_i$ on $H^{1,1}_{\mathrm{prim}}(S_0, \mathbb{R})$ is non-zero, by non-degeneracy of the intersection pairing and $\eta_i \neq 0$. Avoiding their finitely many proper kernels, choose

$$\lambda \in H^{1,1}_{\mathrm{prim}}(S_0, \mathbb{R}) \qquad \lambda \cdot \eta_i \neq 0 \quad (1 \leq i \leq N) \tag{11}$$

By (5), the derivative in the B variables of

$$\begin{aligned} \Psi_1 &: \Lambda_{n,\mathbb{C}} \times B \longrightarrow \mathbb{C}^h \\ \Psi_1(\nu, b) &= \big(\nu \cdot \omega_j(b)\big)_{0 \leq j < h} \end{aligned} \tag{12}$$

is surjective at $(\lambda, 0)$. Applying the holomorphic implicit function theorem on an h-coordinate slice, as in the proof of Theorem 3.10, gives a holomorphic map τ_1 near λ with $\tau_1(\lambda) = 0$ and $\Psi_1(\nu, \tau_1(\nu)) = 0$. Choose non-zero vectors $\nu_m \in \Lambda_n \otimes_{\mathbb{Z}} \mathbb{Q}$ converging to λ in the domain of τ_1, and put $b_m = \tau_1(\nu_m)$. Then $b_m \to 0$ and ν_m is Hodge on S_{b_m}.

Write F^1 for the holomorphic Hodge-filtration subbundle with fibre $H^{2,0}(S_b) \oplus H^{1,1}_{\mathrm{prim}}(S_b)$. By [SB23, Prop. 5.1], the holomorphic quotient F^1/F^2 is the direct image of a line bundle on the relative ramification divisor. Over the ordered chart B, this divisor splits into N copies of B, giving line subbundles of F^1/F^2. The smooth Hodge splitting identifies this quotient with the varying spaces $H^{1,1}_{\mathrm{prim}}(S_b)$ and these line subbundles with the Schiffer image lines. After shrinking B, choose continuous non-zero representatives $\eta_i(b)$ with $\eta_i(0) = \eta_i$. Hence

$$\nu_m \cdot \eta_i(b_m) \longrightarrow \lambda \cdot \eta_i \neq 0$$

For sufficiently large m, an indivisible generator α_m of $\Lambda_n \cap \mathbb{Q}\nu_m$ therefore satisfies the hypotheses of Proposition 3.16 at b_m. Its 1-class Hodge locus contains exact-rank-1 points arbitrarily close to b_m, whose images lie in U. Since U was arbitrary, the assertion follows. ∎

For several imposed classes, it again suffices to detect each additional integral class that is Hodge at 0 and lies outside their rational span by a tangent direction preserving all the imposed classes. We first state the geometric criterion that guarantees generic exact rank, and then construct tuples satisfying it.

Extend Φ_γ complex-linearly to flat classes $\gamma \in \Lambda_{n,\mathbb{C}}$, and write $A_{\gamma,b} = \mathrm{d}\Phi_\gamma(b) \colon T_b B \to \mathbb{C}^h$. For a tuple $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_r)$, put

$$\begin{aligned} \Phi_{\boldsymbol{\alpha}} &= (\Phi_{\alpha_1}, \ldots, \Phi_{\alpha_r}) \\ Z_{\boldsymbol{\alpha}} &= \bigcap_{a=1}^{r} Z_{\alpha_a} \\ \mathcal{A}_{\boldsymbol{\alpha},b} &= \mathrm{d}\Phi_{\boldsymbol{\alpha}}(b) \end{aligned}$$

When $\mathcal{A}_{\boldsymbol{\alpha},0}$ has rank rh, its kernel is the tangent space to $Z_{\boldsymbol{\alpha}}$ at 0. The kernel equality below says that the only primitive $(1,1)$-classes whose period derivatives vanish in every such tangent direction are the linear combinations of the imposed

classes. For integral classes, this is enough to exclude every additional class outside their rational span.

Proposition 3.18. *Let $\alpha_1, \dots, \alpha_r \in \Lambda_n$ be linearly independent classes of type $(1,1)$ on S_0, and put $L_{\mathbb{Q}} = \operatorname{span}_{\mathbb{Q}}\{\alpha_1, \dots, \alpha_r\}$ and $L = \Lambda_n \cap L_{\mathbb{Q}}$. Suppose that $\mathcal{A}_{\boldsymbol{\alpha},0}$ has rank rh and that*

$$\left\{\beta \in H^{1,1}_{\mathrm{prim}}(S_0) : A_{\beta,0}\big|_{\ker \mathcal{A}_{\boldsymbol{\alpha},0}} = 0\right\} = \operatorname{span}_{\mathbb{C}}\{\alpha_1, \dots, \alpha_r\} \tag{13}$$

Then $Z_{\boldsymbol{\alpha}}$ is smooth of complex codimension rh at 0. For every $\beta \in \Lambda_n \smallsetminus L_{\mathbb{Q}}$ that is Hodge at 0, one has

$$\mathrm{d}\Phi_\beta(0)\big|_{T_0 Z_{\boldsymbol{\alpha}}} \neq 0$$

After replacing $Z_{\boldsymbol{\alpha}}$ by a sufficiently small connected representative of its germ, there is a countable union $\mathcal{E}$ of proper closed analytic subsets such that

$$\Lambda_n \cap H^{1,1}(S_b) = L \qquad \operatorname{rk}\mathrm{MW}(S_b/\mathbb{P}^1) = r \qquad (b \in Z_{\boldsymbol{\alpha}} \smallsetminus \mathcal{E})$$

The complement of $\mathcal{E}$ is dense and has full measure for every measure absolutely continuous with respect to smooth volume on $Z_{\boldsymbol{\alpha}}$.

Proof. The rank hypothesis and the holomorphic implicit function theorem give smoothness and $T_0 Z_{\boldsymbol{\alpha}} = \ker \mathcal{A}_{\boldsymbol{\alpha},0}$. Shrink to a connected representative. If an integral class β is Hodge at 0 and its period derivative vanishes on this tangent space, condition (13) gives $\beta = \sum_{a=1}^{r} c_a \alpha_a$ with $c_a \in \mathbb{C}$. Choose an integral basis of Λ_n and an invertible $r \times r$ minor of the coordinate matrix with columns $\alpha_1, \dots, \alpha_r$. Solving the corresponding equations for the c_a shows that every c_a is rational. Hence $\beta \in L_{\mathbb{Q}}$, proving the first-order assertion.

For every $\beta \in \Lambda_n \smallsetminus L_{\mathbb{Q}}$, the subset $Z_{\boldsymbol{\alpha}} \cap Z_\beta$ is closed analytic and proper. If β is not Hodge at 0, it misses the centre; otherwise properness follows from the first-order assertion. Set

$$\mathcal{E} = \bigcup_{\beta \in \Lambda_n \smallsetminus L_{\mathbb{Q}}} \left(Z_{\boldsymbol{\alpha}} \cap Z_\beta\right)$$

As in the proof of Proposition 3.16, countability of the lattice, Baire's theorem, and the measure-zero property of proper analytic subsets give the density and measure assertions.

All the α_a remain Hodge on $Z_{\boldsymbol{\alpha}}$, so every element of L is Hodge there. Outside $\mathcal{E}$, no integral class outside $L_{\mathbb{Q}}$ is Hodge. Thus the primitive integral Hodge lattice is exactly L. The lattice L has rank r and is saturated in Λ_n, meaning that Λ_n / L is torsion-free. The integral Shioda identification therefore gives the asserted Mordell–Weil rank. ∎

It remains to find tuples satisfying (13). As in the 1-class case, we first work with complex coefficient vectors. The following lemma constructs tuples whose stacked period Jacobian has full rank and whose common tangent kernel detects every coefficient vector outside the imposed span. Expressing these conditions by non-vanishing minors will then allow us to pass to real tuples.

Lemma 3.19. *Let $z_1, \dots, z_N$ be distinct complex numbers, and let $h, r \geq 1$ satisfy $rh \leq N$. Assume that either $h = 1$ or $r(h+1) \leq N$. Put $V = (z_i^j)_{0 \leq j < h,\, 1 \leq i \leq N}$ and*

$B_x = V\,\mathrm{diag}(x)$ *for* $x \in \mathbb{C}^N$. *For* $\boldsymbol{x} = (x^1, \ldots, x^r) \in (\mathbb{C}^N)^r$, *set*

$$\mathcal{A}_{\boldsymbol{x}} = \begin{pmatrix} B_{x^1} \\ \vdots \\ B_{x^r} \end{pmatrix} \qquad \mathcal{K}_{\boldsymbol{x}} = \left\{ y \in \mathbb{C}^N : \ker \mathcal{A}_{\boldsymbol{x}} \subseteq \ker B_y \right\}$$

There is a non-empty Zariski-open subset $\mathcal{U} \subset (\mathbb{C}^N)^r$ *such that every* $\boldsymbol{x} \in \mathcal{U}$ *satisfies*

$$\mathrm{rk}\,\mathcal{A}_{\boldsymbol{x}} = rh \qquad \mathcal{K}_{\boldsymbol{x}} = \mathrm{span}_{\mathbb{C}}\{x^1, \ldots, x^r\} \tag{14}$$

Proof. If $h = 1$, the rows of $\mathcal{A}_{\boldsymbol{x}}$ are $x^1, \ldots, x^r$. For every linearly independent tuple, the annihilator of $\ker \mathcal{A}_{\boldsymbol{x}}$ is their row space, giving (14). Such tuples form a non-empty Zariski-open subset because $r \leq N$.

For exact rank, each block must also detect whether an additional period condition vanishes along the simultaneous Hodge locus. We therefore use blocks of size $h+1$: the extra column leaves a one-dimensional kernel, and vanishing on this kernel will force the additional coefficient vector to be proportional to the imposed one on that block.

Suppose $h \geq 2$. Choose disjoint subsets $C_1, \ldots, C_r \subset \{1, \ldots, N\}$ with $|C_a| = h+1$, and let x^a be the indicator vector of C_a. After ordering the columns by these blocks, the non-zero blocks of $\mathcal{A}_{\boldsymbol{x}}$ are the matrices V_{C_a}, each of rank h. Hence $\mathrm{rk}\,\mathcal{A}_{\boldsymbol{x}} = rh$.

For each a, the kernel of V_{C_a} is one-dimensional. Choose a non-zero generator ρ^a. Every coordinate of ρ^a is non-zero: otherwise it would give a non-trivial relation among at most h Vandermonde columns, which are linearly independent. Extend ρ^a by zero outside C_a; the resulting vector belongs to $\ker \mathcal{A}_{\boldsymbol{x}}$.

Let $y \in \mathcal{K}_{\boldsymbol{x}}$. If i lies outside all the blocks, the coordinate vector supported at i belongs to $\ker \mathcal{A}_{\boldsymbol{x}}$. Its image under B_y has first coordinate y_i, so $y_i = 0$. On each block, the condition $B_y \rho^a = 0$ gives

$$V_{C_a}\,\mathrm{diag}(y|_{C_a})\rho^a = 0$$

Since $\ker V_{C_a} = \mathbb{C}\rho^a$, there is a scalar c_a such that $\mathrm{diag}(y|_{C_a})\rho^a = c_a\rho^a$. All coordinates of ρ^a are non-zero, so $y_i = c_a$ for every $i \in C_a$. Thus $y = \sum_a c_a x^a$. Conversely, every linear combination of the x^a belongs to $\mathcal{K}_{\boldsymbol{x}}$, so this tuple satisfies (14).

It remains to show that the two conditions persist on a Zariski-open neighbourhood of this tuple. We express the restriction of B_y to $\ker \mathcal{A}_{\boldsymbol{x}}$ in coordinates, so that the required kernel dimension becomes a matrix-rank condition. Choose a set I of rh columns for which $(\mathcal{A}_{\boldsymbol{x}})_I$ is invertible at the constructed tuple, and put $J = \{1, \ldots, N\} \setminus I$. On the open set where this minor remains invertible, define the linear map $R_{\boldsymbol{x}} : \mathbb{C}^N \to \mathrm{Mat}_{h,N-rh}(\mathbb{C})$ by

$$R_{\boldsymbol{x}}(y) = (B_y)_J - (B_y)_I (\mathcal{A}_{\boldsymbol{x}})_I^{-1} (\mathcal{A}_{\boldsymbol{x}})_J$$

The kernel of $\mathcal{A}_{\boldsymbol{x}}$ consists of the vectors whose coordinates satisfy

$$u_I = -(\mathcal{A}_{\boldsymbol{x}})_I^{-1} (\mathcal{A}_{\boldsymbol{x}})_J u_J$$

Consequently, $\mathcal{K}_{\boldsymbol{x}} = \ker R_{\boldsymbol{x}}$. Full row rank of $\mathcal{A}_{\boldsymbol{x}}$ implies that the x^a are linearly independent. Their span always lies in $\ker R_{\boldsymbol{x}}$, so $\mathrm{rk}\,R_{\boldsymbol{x}} \leq N - r$. At the constructed tuple, equality holds.

After choosing bases for the source and target of the linear map $R_{\boldsymbol{x}}$, its coefficient matrix has entries that are rational functions of $\boldsymbol{x}$, with denominators given by powers of the chosen invertible minor. Non-vanishing of a suitable $(N-r)\times(N-r)$ minor of this coefficient matrix therefore defines a non-empty Zariski-open neighbourhood on which $\operatorname{rk} R_{\boldsymbol{x}} = N-r$. This proves the lemma. ∎

We now prove Theorem 3.14. In the first-order cases, we pull the open set of coefficient tuples back to the real Hodge space and choose a tuple there. We approximate this tuple by rational classes in the full real primitive cohomology and deform the surface so that the approximations become Hodge. We also verify that the kernel condition survives this deformation. The rank-10 case uses the corresponding second-order construction from the next subsection.

Proof of Theorem 3.14. Let $U \subset \mathcal{W}_{\min,n}$ be non-empty and analytically open, and let $H > 0$. By Lemma 3.3, choose a relatively compact marked deformation chart with image in $U \cap \mathcal{W}_n^{\mathrm{sb}}$. Lemma 3.7 gives a non-empty open subset on which no non-zero integral primitive-cohomology Hodge class has $-\gamma^2/2 \le H$. Choose a sufficiently small full marked deformation chart B inside this subset, centred at 0, and use the notation above.

Suppose first that $n \ge 3$ and $r = 10$. Apply Proposition 3.20 inside this chart. It gives an 8-dimensional simultaneous fixed-class Hodge germ defined by 10 independent rational classes, with generic Mordell–Weil rank 10. Replace each rational class by an indivisible integral generator of its rational line. This changes neither the common Hodge locus nor the second-jet kernel condition. After shrinking to a connected representative $Z \subset B$, the chosen height-avoidance condition gives $\widehat{h}(P) > H$ for every non-zero section on every surface over Z. This proves the assertion in this case.

In the remaining cases, either $n = 2$ or $r \le 9$. We use the first-order criterion. Put $W = H^{1,1}_{\mathrm{prim}}(S_0, \mathbb{R})$. The map

$$\kappa \colon W \otimes_{\mathbb{R}} \mathbb{C} \longrightarrow \mathbb{C}^N \qquad \kappa(v) = \big(c_i(\omega_0)(v \cdot \eta_i)\big)_{i=1}^{N}$$

is an isomorphism because the intersection pairing is non-degenerate, the η_i form a basis, and every $c_i(\omega_0)$ is non-zero.

If $n = 2$, then $h = 1$ and $r \le N = 18$. If $n \ge 3$, then

$$r(h+1) = rn \le 9n \le 10n - 2 = N$$

Lemma 3.19 therefore applies. Pulling its non-empty Zariski-open subset back through κ gives a non-empty Zariski-open subset of $(W \otimes_{\mathbb{R}} \mathbb{C})^r$. Since the real form W^r is Zariski dense, choose a real tuple $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_r) \in W^r$ whose image satisfies (14). The tuple is linearly independent. By (5), its simultaneous period Jacobian has rank rh, and the kernel condition translates into

$$\Big\{\beta \in H^{1,1}_{\mathrm{prim}}(S_0) : A_{\beta,0}\big|_{\ker \mathcal{A}_{\boldsymbol{\lambda},0}} = 0\Big\} = \operatorname{span}_{\mathbb{C}}\{\lambda_1, \dots, \lambda_r\}$$

As in the local realization argument, we approximate $\boldsymbol{\lambda}$ by rational classes in the full cohomology space and deform the surface so that these approximations

become Hodge. We must also retain the kernel condition, which will exclude additional persistent Hodge classes outside the span of the imposed classes. Consider

$$\Psi_r : (\Lambda_{n,\mathbb{C}})^r \times B \longrightarrow \mathbb{C}^{rh}$$
$$\Psi_r(\boldsymbol{v}, b) = \big(v_a \cdot \omega_j(b)\big)_{\substack{1\le a\le r\\ 0\le j<h}}$$

Its derivative in the B variables is surjective at $(\boldsymbol{\lambda}, 0)$. Applying the holomorphic implicit function theorem on an rh-coordinate slice, as in the proof of Theorem 3.10, gives a holomorphic map τ_r from a neighbourhood of $\boldsymbol{\lambda}$ to B, with $\tau_r(\boldsymbol{\lambda}) = 0$ and $\Psi_r(\boldsymbol{v}, \tau_r(\boldsymbol{v})) = 0$.

Choose rational tuples $\boldsymbol{v}_m = (v_{1,m}, \dots, v_{r,m}) \in (\Lambda_n \otimes_{\mathbb{Z}} \mathbb{Q})^r$ converging to $\boldsymbol{\lambda}$ in this neighbourhood, and put $b_m = \tau_r(\boldsymbol{v}_m)$. Then $b_m \to 0$, and every $v_{a,m}$ is Hodge on S_{b_m} because it is real and its period pairings vanish. For sufficiently large m, the tuple remains linearly independent and $\mathcal{A}_{\boldsymbol{v}_m, b_m}$ has rank rh.

We check that the kernel condition also persists. For tuples $\boldsymbol{v}$ of Hodge classes at b with $\operatorname{rk} \mathcal{A}_{\boldsymbol{v},b} = rh$, consider the complex-linear map

$$T_{\boldsymbol{v},b} : H^{1,1}_{\mathrm{prim}}(S_b) \longrightarrow \operatorname{Hom}_{\mathbb{C}}\big(\ker \mathcal{A}_{\boldsymbol{v},b}, \mathbb{C}^h\big)$$
$$T_{\boldsymbol{v},b}(\beta) = A_{\beta,b}\big|_{\ker \mathcal{A}_{\boldsymbol{v},b}}$$

The kernel of this map consists of the primitive $(1,1)$-classes whose period equations vanish to first order along the simultaneous Hodge locus. It always contains the imposed classes; we must show that no additional kernel directions appear under a small deformation.

At $(\boldsymbol{\lambda}, 0)$, the map $T_{\boldsymbol{\lambda},0}$ has rank $N - r$. The smooth Hodge splitting gives local continuous frames for its domain spaces, while an invertible rh-column minor of $\mathcal{A}_{\boldsymbol{v},b}$ gives a continuously varying basis of $\ker \mathcal{A}_{\boldsymbol{v},b}$. Thus the matrix entries of $T_{\boldsymbol{v},b}$ vary continuously. A minor witnessing rank $N - r$ at $(\boldsymbol{\lambda}, 0)$ remains non-zero at $(\boldsymbol{v}_m, b_m)$ for sufficiently large m, giving $\operatorname{rk} T_{\boldsymbol{v}_m, b_m} \ge N - r$. On the other hand, the r independent classes $v_{a,m}$ lie in its kernel, giving $\operatorname{rk} T_{\boldsymbol{v}_m, b_m} \le N - r$. Consequently,

$$\ker T_{\boldsymbol{v}_m, b_m} = \operatorname{span}_{\mathbb{C}}\{v_{1,m}, \dots, v_{r,m}\}$$

For each a, let $\alpha_{a,m}$ be an indivisible generator of $\Lambda_n \cap \mathbb{Q} v_{a,m}$. Rescaling the individual classes preserves the simultaneous Hodge locus, the rank of its period Jacobian, and the kernel condition. Proposition 3.18, applied at b_m, therefore gives a smooth connected local simultaneous Hodge locus Z of dimension $N - rh$ on which the Mordell–Weil rank is exactly r outside a countable union of proper closed analytic subsets.

Shrink Z inside B. By the choice of B and the integral Shioda identification, every non-zero section on every surface parametrized by Z has canonical height greater than H. The complement of the exceptional analytic subsets is non-empty and maps into U, proving exact-rank-r density as well as the stronger assertion. ∎

For $n \ge 3$, the range $r \le 9$ obtained by the first-order part of the proof of Theorem 3.14 is the dimensional limit of the complex first-order criterion (13). Indeed, the restriction map $T_{\boldsymbol{v},b}$ in the preceding proof must have rank $N - r$, while its target has dimension $h(N - rh)$. Thus, for $h > 1$,

$$N - r \le h(N - rh) \quad \Longleftrightarrow \quad r(h+1) \le N$$

Since $h = n-1$ and $N = 10n-2$, this is equivalent to $r \le 9$. Lemma 3.19 attains this range.

The failure of the complex first-order criterion at rank 10 can also be seen directly. At a transverse 10-class point, the imposed Hodge locus has tangent dimension 8, so the restriction map $T_{\boldsymbol{\nu},b}$ has source dimension $N = 10h+8$ and target dimension $8h$. Consequently,

$$\dim_{\mathbb{C}} \ker T_{\boldsymbol{\nu},b} \ge N - 8h = 2h+8 > 10 \qquad (h \ge 2)$$

Thus its kernel contains complex primitive $(1,1)$-classes outside the span of the 10 imposed classes. These additional classes need not be rational, and vanishing of their period differentials along the tangent space does not imply that they remain Hodge along the actual germ. Their existence therefore defeats the direct first-order criterion without forcing additional Mordell–Weil sections.

The next subsection uses second derivatives to obtain generic exact rank 10 for every $n \ge 3$.

3.7. **Second-order period equations and exact rank 10.** The first-order criterion cannot isolate the 10 imposed classes when $n \ge 3$: some additional complex $(1,1)$-classes necessarily have vanishing period derivatives along the 8-dimensional germ. To obtain exact rank 10, we will detect these remaining classes by second derivatives. Our goal is to construct a germ on which the only primitive $(1,1)$-classes whose restricted periods vanish through second order are combinations of the 10 imposed classes. An additional rational class then cannot remain Hodge throughout the germ.

Keep $n \ge 3$, $h = n-1$, and $N = 10n-2$. On an N-dimensional marked deformation chart with a holomorphic frame $\omega_0, \dots, \omega_{h-1}$ of F^2, use the period sections $\Phi_\beta = (\beta \cdot \omega_j)_{0 \le j < h}$ from the preceding subsection. If Z is a smooth local submanifold through b, write $j_b^2(\Phi_\beta|_Z)$ for its Taylor jet through degree two. Its vanishing means that the value and all first and second derivatives along Z vanish at b.

Proposition 3.20. *Let $n \ge 3$. For every non-empty analytic open subset $U \subset \mathcal{W}_n^{\mathrm{sb}}$, there are a marked deformation chart $B \to \mathcal{W}_n^{\mathrm{sb}}$ and a connected smooth 8-dimensional simultaneous fixed-class Hodge locus $Z \subset B$, with image in U, cut out transversely by the period equations of 10 independent classes $\alpha_1, \dots, \alpha_{10} \in \Lambda_{n,\mathbb{Q}}$, such that*

$$\left\{\beta \in H^{1,1}_{\mathrm{prim}}(S_b) : j_b^2(\Phi_\beta|_Z) = 0\right\} = L_{\mathbb{C}} \qquad (b \in Z) \tag{15}$$

where $L_{\mathbb{Q}} = \operatorname{span}_{\mathbb{Q}}\{\alpha_1, \dots, \alpha_{10}\}$ and $L_{\mathbb{C}} = L_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C}$. Outside a countable union of proper closed analytic subsets of Z, the Mordell–Weil rank is exactly 10. If $\beta \in \Lambda_{n,\mathbb{Q}} \smallsetminus L_{\mathbb{Q}}$ is Hodge at $b \in Z$, then $\Phi_\beta|_Z$ has vanishing order one or two at b.

We first compute the second derivative of a period section along the common zero locus of the imposed periods.

Let $\alpha_1, \dots, \alpha_{10} \in H^{1,1}_{\mathrm{prim}}(S_b)$ be a possibly complex tuple for which

$$J = d\Phi_{\boldsymbol{\alpha},b} \colon T_b B \longrightarrow \mathbb{C}^{10h}$$

is surjective, where $\Phi_{\boldsymbol{\alpha}} = (\Phi_{\alpha_1}, \dots, \Phi_{\alpha_{10}})$. Their common period-zero germ Z is smooth of dimension 8, with $K = T_b Z = \ker J$.

The second derivative must be taken along Z, whose embedding in B contributes a normal correction. Choose a right inverse R of J, so $JR = \mathrm{id}$. If $c(t)$ is a holomorphic curve in Z with $c(0) = b$ and $c'(0) = u \in K$, differentiating $\Phi_{\boldsymbol{\alpha}}(c(t)) = 0$ twice gives

$$Jc''(0) = -d^2\Phi_{\boldsymbol{\alpha},b}(u,u)$$

Hence $c''(0) \equiv -R\,d^2\Phi_{\boldsymbol{\alpha},b}(u,u) \pmod{K}$.

Suppose that $\beta \in H^{1,1}_{\mathrm{prim}}(S_b)$ and $d\Phi_{\beta,b}|_K = 0$. The chain rule for $\Phi_\beta(c(t))$, followed by polarization, gives the Hessian of the restricted period section:

$$\mathcal{Q}_\beta(u,v) = d^2\Phi_{\beta,b}(u,v) - d\Phi_{\beta,b}R\,d^2\Phi_{\boldsymbol{\alpha},b}(u,v) \qquad u,v \in K \tag{16}$$

The difference between two right inverses takes values in K, which is annihilated by $d\Phi_{\beta,b}$, so the expression is independent of R. Since the period value and its restricted first differential vanish, this Hessian is intrinsic, with values in $(F_b^2)^*$. Its vanishing is independent of the coordinates and the chosen frame.

We establish the required non-vanishing at the weighted Fermat surface S_0 below, which lies outside $\mathcal{W}_n^{\mathrm{sb}}$. Its diagonal symmetries distinguish the Hodge types of the residue classes and make the period derivatives explicit.

Lemma 3.21. *Let $n \geq 3$, and let S_0 be the minimal resolution of the hypersurface*

$$F_0 = Y^2 + X^3 + s^D + t^D = 0 \qquad D = 6n$$

in the coarse weighted projective space $\mathbb{P}(1,1,2n,3n)$. There exist an N-dimensional marked local deformation chart $(B,0)$ centred at S_0 and 10 linearly independent complex classes $\alpha_1,\dots,\alpha_{10} \in H^{1,1}_{\mathrm{prim}}(S_0)$ such that

$$J = d\Phi_{\boldsymbol{\alpha},0} \colon T_0B \longrightarrow \mathbb{C}^{10h}$$

is surjective. Their common period-zero germ $(Z,0)$ is smooth of dimension 8 and satisfies

$$\left\{\beta \in H^{1,1}_{\mathrm{prim}}(S_0) : j_0^2(\Phi_\beta|_Z) = 0\right\} = \mathrm{span}_{\mathbb{C}}\{\alpha_1,\dots,\alpha_{10}\}$$

Proof. The affine cone is smooth away from its vertex, so the hypersurface is quasi-smooth. Its only coarse singular point lies on $s = t = 0$. In the chart $X = 1$, choose one of the two non-zero values of Y. Since $\partial F_0/\partial Y \neq 0$, the implicit function theorem eliminates Y; the residual stabilizer is μ_n, acting on (s,t) by scalar multiplication. The singularity is therefore of type $\frac{1}{n}(1,1)$.

The same calculation with the coefficients as parameters gives a local product of this quotient singularity with the parameter space. Resolving this product gives a simultaneous resolution, whose exceptional curve is the zero section and has self-intersection $-n$. After shrinking, the resolved family is smooth. Its central member S_0 is an elliptic surface of height n with $6n$ type-II fibres, since $s^{6n} + t^{6n}$ has simple zeros. In particular, this seed does not lie in $\mathcal{W}_n^{\mathrm{sb}}$; we will pass to that locus after establishing the required non-vanishing.

The primitive residue cohomology identifies with $\langle F, O\rangle^\perp$ on the resolution. Indeed, the exceptional class is O, and the pullback of the rational weighted hyperplane class is $F + \frac{1}{n}O$. Orthogonality to these two classes is therefore equivalent to

orthogonality to F and O. The quasi-smooth residue description recalled in [Klo07, §8, p. 310 and Lem. 8.2] applies. The Jacobian ring is

$$R_{F_0} = \mathbb{C}[s,t,X]/(s^{D-1}, t^{D-1}, X^2)$$

and its relevant graded pieces are

$$T = (R_{F_0})_D \qquad H^{2,0} = (R_{F_0})_{n-2}$$
$$H^{1,1}_{\text{prim}} = (R_{F_0})_{7n-2} \qquad H^{0,2} = (R_{F_0})_{13n-2}$$

Here T is the tangent space to the local deformation slice obtained by fixing the coefficients of $s^D, s^{D-1}t, st^{D-1}, t^D$.

Let Ω be the contraction of $ds \wedge dt \wedge dX \wedge dY$ with the weighted Euler vector field $s\partial_s + t\partial_t + 2nX\partial_X + 3nY\partial_Y$. Take

$$\omega_j = \operatorname{Res} \frac{s^j t^{h-1-j}\Omega}{F_0} \qquad 0 \le j < h$$

and let U_a, V_a be the classes dual under the intersection pairing to the following two families of residues:

$$\operatorname{Res} \frac{s^a t^{7h+5-a}\Omega}{F_0^2} \qquad (h+1 \le a \le 6h+4)$$
$$\operatorname{Res} \frac{Xs^a t^{5h+3-a}\Omega}{F_0^2} \qquad (0 \le a \le 5h+3)$$

These residues lie in $H^{1,1}_{\text{prim}}(S_0)$ itself. A reduced monomial $s^a t^b X^c$ has character $(a+1, b+1, c+1, 1)$ under the diagonal Fermat automorphisms, with entries taken modulo $(D, D, 3, 2)$. These characters are distinct across the three reduced residue bases, so each eigenspace is one-dimensional and belongs to a single Hodge summand.

Use the monomials

$$B_a = s^a t^{D-a} \quad (2 \le a \le D-2) \qquad A_a = Xs^a t^{4n-a} \quad (0 \le a \le 4n)$$

as coefficient directions on the slice. For fixed g of degree $n-2$, the residues $\omega_g = \operatorname{Res}(g\Omega/F)$ extend holomorphically over this linear family. Differentiation in coefficient directions G, H gives

$$\partial_G \omega_g = -\operatorname{Res} \frac{gG\Omega}{F_0^2}$$
$$\partial_G \partial_H \omega_g = 2\operatorname{Res} \frac{gGH\Omega}{F_0^3} \tag{17}$$

Consequently, for $\beta = \sum u_i U_i + \sum v_i V_i$,

$$\big(d\Phi_\beta(B_a)\big)_j = -u_{a+j} \qquad \big(d\Phi_\beta(A_a)\big)_j = -v_{a+j} \tag{18}$$

where coordinates outside the indicated ranges are understood to be zero.

Choose

$$(\alpha_1, \dots, \alpha_5) = (U_{h+1}, U_{2h+1}, U_{3h+2}, U_{4h+2}, U_{5h+3})$$
$$\alpha_6 = U_{6h+4} + U_{3h}$$
$$(\alpha_7, \dots, \alpha_{10}) = (V_h, V_{2h+1}, V_{3h+1}, V_{4h+3})$$

and put $L_{\mathbb{C}} = \operatorname{span}_{\mathbb{C}}\{\alpha_1, \ldots, \alpha_{10}\}$. This is an auxiliary complex tuple: Z denotes its period-zero germ, and no Hodge interpretation away from the central point is needed at this stage.

Formula (18) shows that J has rank $10h$. Set $q = B_{5h+6} - B_{2h+2}$, $r = B_{4h+3}$, and $w = A_0$. The kernel of J has basis

$$q,\ r,\ B_{5h+4},\ w,\ A_{h+1},\ A_{3h+2},\ A_{3h+3},\ A_{4h+4}$$

Let G be the subspace spanned by

$$U_{h+2}, \ldots, U_{2h} \qquad U_{3h+3}, \ldots, U_{4h+1} \qquad V_{2h+2}, \ldots, V_{3h}$$

To see which classes have vanishing first derivative on K, the directions r and B_{5h+4} first give

$$u_i = 0 \qquad (4h+3 \le i \le 5h+2)$$
$$u_i = 0 \qquad (5h+4 \le i \le 6h+3)$$

The direction q then gives

$$u_i = 0 \quad (2h+2 \le i \le 3h-1) \qquad u_{3h+1} = 0 \qquad u_{3h} = u_{6h+4}$$

The last equality accounts for the mixed class α_6. The five A-directions leave only the four imposed V-coordinates and the coordinates $v_{2h+2}, \ldots, v_{3h}$. Consequently,

$$\left\{\beta \in H^{1,1}_{\mathrm{prim}}(S_0) : d\Phi_\beta|_K = 0\right\} = L_{\mathbb{C}} \oplus G \tag{19}$$

Thus the first derivatives leave $3(h-1)$ additional directions. We will show that three families of restricted Hessian components detect all of them.

In reducing the second expression in (17), a numerator exponent $e_s \ge D$ contributes $(e_s - D + 1)/D$ times the pole-two residue obtained by replacing s^{e_s} with $s^{e_s - D}$; the same rule applies to t. The factor 2 in the second derivative cancels the factor $1/2$ in Griffiths pole reduction. An exponent equal to $D-1$ gives zero. A factor X^2 also gives zero, since division by $\partial_X F_0 = 3X^2$ leaves an X-independent numerator. By the Fermat-character argument above, the final pole-one and reduced pole-three terms are pure of types $(2,0)$ and $(0,2)$, respectively, so they pair trivially with every $\beta \in H^{1,1}_{\mathrm{prim}}(S_0)$.

For $\beta = \sum_i u_i U_i + \sum_i v_i V_i$, direct reduction gives, for $0 \le j < h$,

$$\begin{aligned} D\big(d^2\Phi_\beta(q,q)\big)_j &= (4h+7+j)u_{4h+6+j} - 2(h+3+j)u_{h+2+j} \\ &\quad + (3h+2-j)u_{4h+4+j} \\ D\big(d^2\Phi_\beta(r,q)\big)_j &= (3h+4+j)u_{3h+3+j} \\ D\big(d^2\Phi_\beta(q,w)\big)_j &= -(3h+2-j)v_{2h+2+j} \end{aligned}$$

The three terms in the first identity come from B_{5h+6}^2, $-2B_{5h+6}B_{2h+2}$, and B_{2h+2}^2, respectively. In the second identity, the contribution from $B_{4h+3}B_{2h+2}$ either vanishes or reduces to pole one. In the third, the contribution from $B_{5h+6}A_0$ either vanishes or is a reduced pole-three term.

We must now subtract the normal correction in (16). Let $e_{a,j}$ denote the standard coordinate vector of $\mathbb{C}^{10h}$ corresponding to the jth period equation for α_a. Formula

(18) gives

$$\begin{aligned} JB_{h+2} &= -e_{2,h-1} & JB_{3h+3} &= -e_{4,h-1} \\ JA_{2h+2} &= -e_{9,h-1} & JB_{4h+4} &= -e_{5,h-1} \\ JB_{4h+6} &= -e_{5,h-3} \quad (h \geq 3) \end{aligned}$$

The target coordinate vectors are distinct, so we may choose a right inverse satisfying

$$\begin{aligned} Re_{2,h-1} &= -B_{h+2} & Re_{4,h-1} &= -B_{3h+3} \\ Re_{9,h-1} &= -A_{2h+2} & Re_{5,h-1} &= -B_{4h+4} \\ Re_{5,h-3} &= -B_{4h+6} \quad (h \geq 3) \end{aligned}$$

Put $C_h = B_{4h+6}$ for $h \geq 3$ and $C_2 = 0$. Substituting the 10 imposed classes into the ambient second derivatives yields

$$\begin{aligned} DRd^2\Phi_{\boldsymbol{\alpha}}(q,q) &= (4h+4)B_{h+2} - (2h+3)B_{4h+4} \\ &\quad -(5h+4)C_h \\ DRd^2\Phi_{\boldsymbol{\alpha}}(r,q) &= -(4h+3)B_{3h+3} \\ DRd^2\Phi_{\boldsymbol{\alpha}}(q,w) &= (2h+3)A_{2h+2} \end{aligned}$$

For the first identity, the non-zero components of $D\,d^2\Phi_{\boldsymbol{\alpha}}(q,q)$ are $-(4h+4)e_{2,h-1}$ and $(2h+3)e_{5,h-1}$, together with $(5h+4)e_{5,h-3}$ when $h \geq 3$. The other two identities have the single non-zero components $(4h+3)e_{4,h-1}$ and $-(2h+3)e_{9,h-1}$, respectively.

Now restrict to $\beta \in G$. Its coordinate support gives $d\Phi_\beta(B_{4h+4}) = d\Phi_\beta(C_h) = 0$. Combining the ambient derivatives and their normal corrections, we obtain, for $0 \leq j \leq h-2$,

$$\begin{aligned} D\mathcal{Q}_{\beta,j}(q,q) &= \big[-2(h+3+j)+4h+4\big]u_{h+2+j} \\ &= 2(h-1-j)u_{h+2+j} \\ D\mathcal{Q}_{\beta,j}(r,q) &= \big[3h+4+j-(4h+3)\big]u_{3h+3+j} \\ &= -(h-1-j)u_{3h+3+j} \\ D\mathcal{Q}_{\beta,j}(q,w) &= \big[-(3h+2-j)+(2h+3)\big]v_{2h+2+j} \\ &= -(h-1-j)v_{2h+2+j} \end{aligned} \tag{20}$$

These $3(h-1)$ components give a diagonal matrix on the displayed basis of G, with determinant

$$\frac{2^{h-1}((h-1)!)^3}{D^{3(h-1)}} \neq 0$$

Every class in the first-order kernel has the form $\ell + g$ with $\ell \in L_{\mathbb{C}}$ and $g \in G$. The imposed period sections vanish identically on Z, so all jets of ℓ vanish. The diagonal calculation forces $g = 0$ whenever the restricted second jet vanishes. Together with (19), this proves the asserted second-jet kernel identity. ∎

Proof of Proposition 3.20. Let S_0 and $\boldsymbol{\alpha}$ be the Fermat surface and complex tuple supplied by Lemma 3.21.

On a chart where a fixed $10h$-column minor of J is non-zero, a basis of $\ker J$ and a right inverse of J depend rationally on the classes α_a. Using the remaining 8 coordinates, write the simultaneous period-zero germ as an implicit graph. The

first and second derivatives of this graph, and hence a matrix for the full restricted second jet, depend rationally on the tuple on this minor chart. Clearing the denominators makes the required non-vanishing polynomial in the tuple. The real form $H^{1,1}_{\mathrm{prim}}(S_0,\mathbb{R})^{10}$ is Zariski dense in its complexification, so there is a real tuple with the same two rank conditions. Both conditions are open in the real-analytic Hodge bundle. Since $\mathcal{W}_n^{\mathrm{sb}}$ is dense, they hold for a nearby real tuple at a simply-branched surface.

To reach every open set, let $T^{\mathrm{sb}} \subset V_n$ be the open coefficient locus parametrizing $\mathcal{W}_n^{\mathrm{sb}}$. It is a non-empty Zariski-open subset of the coefficient space and is connected. Over it, consider the space $\mathcal{T}$ of complex 10-tuples in primitive $(1,1)$-cohomology for which J is surjective. Each fibre is a non-empty complex Zariski-open subset of a vector space by Lemma 3.8, and is connected. Since $\mathcal{T}$ is open in the corresponding real-analytic vector bundle, its projection is open and $\mathcal{T}$ is connected. The jet condition is computed on local deformation slices; pulling it back to the coefficient space leaves its kernel unchanged, since the four coordinate-change directions are isoperiodic.

At $(b,\boldsymbol{\alpha}) \in \mathcal{T}$, the full restricted second-jet map on $H^{1,1}_{\mathrm{prim}}(S_b)$ has a kernel containing the 10-dimensional imposed span. In local frames it is represented by a real-analytic matrix. Failure of (15) is the vanishing of all its $(N-10)\times(N-10)$ minors, defining a real-analytic rank locus in $\mathcal{T}$. Our calculation and perturbation show that this locus is proper, so it has empty interior.

If a non-empty open subset of T^{sb} admitted no suitable tuple, its inverse image in $\mathcal{T}$ would be a non-empty open subset of this failure locus, a contradiction. Thus the base points admitting suitable complex tuples form an open dense set. At each such point, the same polynomial argument gives a real tuple.

Fix a suitable real tuple over a point of the prescribed open set U, and choose a marked deformation chart B around that point with image in U. Approximate this tuple by rational classes in the full real primitive cohomology and apply the parameter implicit function theorem as in Theorem 3.10. The rational classes become Hodge on nearby surfaces, and the two rank conditions persist by openness. Denote the resulting classes by $\alpha_1,\dots,\alpha_{10}$. Their common Hodge germ is cut out transversely and has dimension 8. After shrinking to a connected representative Z, openness ensures that (15) holds at every point of Z, and its image remains in U.

For these rational classes, put $L_{\mathbb{Q}} = \operatorname{span}_{\mathbb{Q}}\{\alpha_1,\dots,\alpha_{10}\}$ and $L_{\mathbb{C}} = L_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{C}$. For each $\beta \in \Lambda_{n,\mathbb{Q}} \smallsetminus L_{\mathbb{Q}}$, the subset $Z \cap \Phi_\beta^{-1}(0)$ is closed analytic in Z. If Φ_β vanished identically on Z, then β would be Hodge at the base point because it is real, and its second jet would vanish. Equation (15) would give $\beta \in L_{\mathbb{C}} \cap \Lambda_{n,\mathbb{Q}} = L_{\mathbb{Q}}$, a contradiction. These subsets are therefore proper. Countability, Baire's theorem, and the integral Shioda identification give exact rank 10 outside their union.

Finally, if an additional rational class is Hodge at $b \in Z$, equation (15) prevents its first and second restricted derivatives from both vanishing there. Its period section therefore has vanishing order one or two. ∎

The Hessian in (16) depends on the second derivative of the period map; the curvature of F^2 is determined by its first differential. The jet condition is intrinsic

and therefore applies in Schiffer coordinates as well as Weierstrass coefficient coordinates. Together with height avoidance, the proposition supplies the rank-10 case of Theorem 3.14.

3.8. **Section moduli and comparison with Hodge loci.** In this subsection, we interpret the realization theorem in terms of moduli of elliptic surfaces with chosen sections. We first express the canonical height of a section as the degree of a line bundle pulled back from $\overline{\mathcal{M}}_{1,2}$. We then compare deformations of the sections with the corresponding period conditions.

Write $\mathcal{Y} = \overline{\mathcal{M}}_{1,1} \simeq \mathcal{P}(4,6)$ and $\mathcal{X} = \overline{\mathcal{M}}_{1,2}$. Inchiostro identifies $\mathcal{X}$ with the $(4,6)$-weighted blowup of $\mathcal{P}(2,3,4)$ at $[1:1:0]$ [Inc22, Thms. 1.1 and 2.10]. Let

$$\phi : \mathcal{X} \longrightarrow \mathcal{Y} \qquad \pi : \mathcal{X} \longrightarrow \mathcal{P}(2,3,4)$$

be the universal stable one-pointed curve and the weighted blowdown, respectively. All weighted projective spaces here are stacks. The exceptional divisor D parametrizes curves whose two markings lie on a rational tail; under the universal-curve identification, it is the zero section. We write $\lambda = \mathcal{O}_{\mathcal{Y}}(1)$, $\mathcal{L}_{1,2} = \phi^*\lambda$, and $\xi = \pi^*\mathcal{O}_{\mathcal{P}(2,3,4)}(1)$. Throughout this subsection, we work on the all-I_1 locus and write $f : S \to \mathbb{P}^1$ for the elliptic surface fibration.

Proposition 3.22. *The line bundles on $\mathcal{X}$ satisfy*

$$\xi \simeq \mathcal{L}_{1,2} \otimes \mathcal{O}_{\mathcal{X}}(D) \qquad \omega_{\mathcal{X}} \simeq \mathcal{L}_{1,2}^{\otimes(-9)} \tag{21}$$

Let $f : S \to \mathbb{P}^1$ represent a point of $\mathcal{W}_n^{\mathrm{I}_1}$, with classifying map φ_S and fundamental line bundle $\mathcal{L}$. A section $P \neq O$ determines a lift $f_P : \mathbb{P}^1 \to \mathcal{X}$ of φ_S. If $C_P = P^*O$ is the divisor where P meets O, then

$$f_P^*\mathcal{L}_{1,2} \simeq \mathcal{L} \qquad f_P^*D = C_P \qquad (\pi \circ f_P)^*\mathcal{O}(1) \simeq \mathcal{L}(C_P) \tag{22}$$

In particular,

$$\deg f_P^*\mathcal{L}_{1,2} = n \qquad \widehat{h}(P) = \deg f_P^*\xi = n + (P \cdot O) \geq n \tag{23}$$

Proof. We first compute the line bundles in (21). In weighted coordinates $[x : y : A]$ on $\mathcal{P}(2,3,4)$, the rational map to $\mathcal{Y}$ is given by A and $y^2 - x^3 - Ax$, of weights 4 and 6. Their pullbacks along π have exceptional factors of orders 4 and 6. Dividing by these factors gives sections of the fourth and sixth powers of $\xi \otimes \mathcal{O}_{\mathcal{X}}(-D)$. These sections, together with this underlying line bundle, define ϕ. Hence $\mathcal{L}_{1,2} \simeq \xi \otimes \mathcal{O}_{\mathcal{X}}(-D)$ in the integral Picard group of $\mathcal{X}$.

The canonical bundle of $\mathcal{P}(2,3,4)$ is $\mathcal{O}(-9)$, and the $(4,6)$-weighted blowup has discrepancy $4+6-1=9$. Hence

$$\omega_{\mathcal{X}} \simeq \xi^{\otimes(-9)} \otimes \mathcal{O}_{\mathcal{X}}(9D) \simeq \mathcal{L}_{1,2}^{\otimes(-9)}$$

which proves the second identity.

We now construct f_P. Since all singular fibres have type I_1, the surface S is the pullback of the universal stable one-pointed curve along φ_S. The section P therefore determines a lift to $\mathcal{X}$. A section cannot pass through a fibre node: in the local equation $xy = t$, the conditions $x(0) = y(0) = 0$ would imply $\mathrm{ord}_0(xy) \geq 2$.

At a point where P meets O, the two-pointed stabilization inserts a rational tail carrying the two markings. Since D is the universal zero section, its pullback is P^*O

as an effective Cartier divisor, with the collision multiplicities included. Together with $\varphi_S^*\lambda \simeq \mathcal{L}$ and the first identity in (21), this proves (22). The map f_P lands in $\mathcal{M}_{1,2}$ precisely where the fibre is smooth and the two sections are distinct.

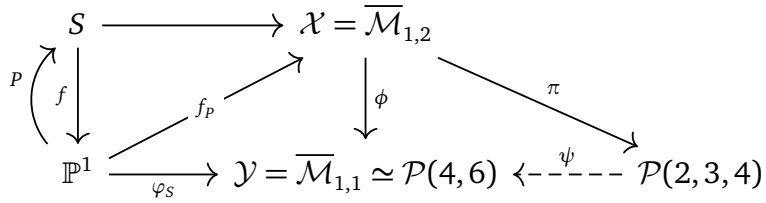
It remains to identify the degree with the canonical height. Put $d = n + (P \cdot O)$. The class $\gamma_P = [P] - [O] - d[F]$ is orthogonal to F because $(P-O)\cdot F = 0$ and $F^2 = 0$. The coefficient d is forced by orthogonality to O, since $(P-O)\cdot O = n + P\cdot O = d$ and $F \cdot O = 1$. Since $P^2 = O^2 = -n$ and $P \cdot F = O \cdot F = 1$, we have $\gamma_P^2 = -2d$. All fibres are irreducible, so γ_P is Shioda's projection of P. Our normalization gives $\widehat{h}(P) = -\gamma_P^2/2 = d$, while (22) gives $\deg f_P^*\xi = d$. Finally, $P \neq O$ implies $P \cdot O \geq 0$, proving (23). ∎

We can also describe the map to $\mathcal{P}(2,3,4)$ directly from the Weierstrass equation. Write $P = (x_P, y_P)$ in $y^2 = x^3 + a_4 x + a_6$, let s be the canonical section of $\mathcal{O}(C_P)$, and put $\mathcal{L}_P = \mathcal{L}(C_P)$. Then

$$X = s^2 x_P \in H^0(\mathcal{L}_P^{\otimes 2})$$
$$Y = s^3 y_P \in H^0(\mathcal{L}_P^{\otimes 3})$$
$$A' = s^4 a_4 \in H^0(\mathcal{L}_P^{\otimes 4})$$

satisfy $Y^2 - X^3 - A'X = s^6 a_6$ and define $\pi \circ f_P = [X : Y : A']$. To check that these sections have no common zero, consider first a collision of multiplicity k. The coordinates x_P and y_P have poles of orders $2k$ and $3k$, so X and Y are non-zero there and the image is $[1 : 1 : 0]$. Away from the collisions, a common zero of X, Y, A' would imply $a_4 = a_6 = 0$, contrary to the all-I_1 hypothesis. Thus $(\mathcal{L}_P, X, Y, A')$ is a basepoint-free $(2,3,4)$-weighted linear series whose underlying line bundle has degree $\deg \mathcal{L}_P = n + (P \cdot O) = \widehat{h}(P)$, in the terminology of [BPS24, Def. 2.9 and Prop. 2.10].

The surface and its chosen section fit into the following commutative diagram:



The dashed arrow is the rational map $\psi([x : y : A]) = [A : y^2 - x^3 - Ax]$, whose indeterminacy at $[1 : 1 : 0]$ is resolved by the $(4,6)$-weighted blowup π. Thus the right triangle commutes wherever ψ is defined. The left square is Cartesian, and f_P is the composite of the section P with the upper horizontal map. The morphism π contracts the zero-section divisor $D \simeq \mathcal{Y}$, whose pullback $f_P^* D = C_P$ records the intersections of P with O. The two line bundles on $\mathcal{X}$ measure the two heights:

$$f_P^*\mathcal{L}_{1,2} \simeq \mathcal{L} \qquad \deg f_P^*\mathcal{L}_{1,2} = n$$
$$f_P^*\xi \simeq \mathcal{L}(C_P) \qquad \deg f_P^*\xi = \widehat{h}(P)$$

Thus the projection to $\mathcal{Y}$ records the Faltings height of the surface, while the composite with π records the canonical height of the chosen section. The exceptional divisor measures their difference:

$$\deg f_P^*\mathcal{O}_{\mathcal{X}}(D) = P \cdot O = \widehat{h}(P) - n$$

We now construct the stack $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$ introduced in the Introduction. Its objects are height n all-I_1 Jacobian elliptic surfaces with ordered non-zero sections $P_1,\dots,P_r$ of canonical heights $d_1,\dots,d_r$, with the base considered up to isomorphism. The sections determine a map to the fibre product

$$\mathcal{X}^{(r)} = \underbrace{\mathcal{X}\times_{\mathcal{Y}}\cdots\times_{\mathcal{Y}}\mathcal{X}}_{r\text{ factors}}$$

Conversely, consider a map $g=(f_1,\dots,f_r)\colon \mathbb{P}^1\to\mathcal{X}^{(r)}$ with common projection $\varphi\colon\mathbb{P}^1\to\mathcal{Y}$. We require $\deg\varphi^*\lambda=n$, $\deg f_a^*\xi=d_a$, and the pullback of the nodal boundary along φ to be a reduced divisor. Pulling back the universal curve along φ then gives the elliptic fibration, and the lifts f_a give its sections. The positivity of d_a excludes the zero section, since $\xi|_D\simeq\mathcal{O}_D$. These constructions commute with base change. Thus $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$ is the corresponding open locus of maps of the specified degrees, modulo PGL_2. The fibre product allows the sections P_a to meet.

Forgetting the additional sections defines a morphism

$$\rho\colon \mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}\longrightarrow\mathcal{W}^{\mathrm{I}_1}_n$$

Its source and target have finite stabilizers; automorphisms are required to preserve the zero section. The degrees $d_a=\deg f_a^*\xi$ are locally constant in families, so fixing $\mathbf{d}$ does not change the local deformation germ of a tuple.

Proposition 3.23. *The stack $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$ is an algebraic Deligne–Mumford stack of finite type over $\mathbb{C}$, and the forgetful morphism ρ is representable. Let $(S,P_1,\dots,P_r)$ be a point of $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$, and let $B\to\mathcal{W}^{\mathrm{I}_1}_n$ be a simply-connected analytic étale local deformation chart through $[S]$. Choose a marking and a lift of the tuple over the central point b_0. Put $\gamma_a=[P_a]-[O]-d_a[F]$, and define*

$$\Phi(b)=\big(\gamma_a\cdot\omega_j(b)\big)_{a,j}\qquad 1\le a\le r,\quad 0\le j\le n-2$$

where $\omega_0(b),\dots,\omega_{n-2}(b)$ is a local holomorphic frame of $H^{2,0}(S_b)$. If $d\Phi$ has rank $r(n-1)$ at b_0, then $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$ is smooth at the tuple, of dimension

$$10n-2-r(n-1) \tag{24}$$

After base change to B, the morphism ρ identifies the germ at the chosen lift with $(\Phi^{-1}(0),b_0)$.

Proof. We first establish the algebraic moduli construction. Over a coefficient atlas for $\mathcal{W}^{\mathrm{I}_1}_n$, the universal surface has a relatively ample divisor $A=3O+cF$ for an integer $c>3n$. A section P_a of canonical height d_a has Hilbert polynomial

$$\chi\big(\mathcal{O}_{P_a}(mA)\big)=\big(3(d_a-n)+c\big)m+1$$

Thus tuples of sections of the specified heights form an open locus in a product of relative Hilbert schemes with fixed Hilbert polynomials. This construction descends to $\mathcal{W}^{\mathrm{I}_1}_n$, proving that $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$ is algebraic of finite type and that ρ is representable. Since $\mathcal{W}^{\mathrm{I}_1}_n$ is Deligne–Mumford, so is $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$.

We next compute the deformations of the corresponding map $g=(f_{P_1},\dots,f_{P_r})\colon\mathbb{P}^1\to\mathcal{X}^{(r)}$. Each component lies in the smooth locus of ϕ. This includes collisions with

O, which correspond to smooth points of the original one-pointed fibre under the universal-curve identification. Thus $\mathcal{X}^{(r)}$ is smooth in a neighbourhood of $g(\mathbb{P}^1)$.

Since $P_a \simeq \mathbb{P}^1$, $K_S = (n-2)F$, and $F \cdot P_a = 1$, adjunction gives $P_a^2 = -2 - K_S \cdot P_a = -n$. Thus $N_{P_a/S} \simeq \mathcal{O}_S(P_a)|_{P_a} \simeq \mathcal{O}_{\mathbb{P}^1}(-n)$. Along a section, the vertical tangent line of $f : S \to \mathbb{P}^1$ identifies with the section's normal bundle. Since S is pulled back from the universal curve, base change therefore identifies $f_{P_a}^* T_{\mathcal{X}/\mathcal{Y}}$ with $N_{P_a/S}$. Together with $\varphi_S^* T_{\mathcal{Y}} \simeq \mathcal{L}^{\otimes 10} \simeq \mathcal{O}_{\mathbb{P}^1}(10n)$, the relative tangent sequence gives, with $E = g^* T_{\mathcal{X}^{(r)}}$,

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^1}(-n)^{\oplus r} \longrightarrow E \longrightarrow \mathcal{O}_{\mathbb{P}^1}(10n) \longrightarrow 0 \tag{25}$$

Taking cohomology gives

$$\begin{aligned} 0 \longrightarrow H^0(E) \longrightarrow H^0(\mathcal{O}_{\mathbb{P}^1}(10n)) \xrightarrow{\delta} \bigoplus_{a=1}^{r} H^1(N_{P_a/S}) \\ \longrightarrow H^1(E) \longrightarrow 0 \end{aligned} \tag{26}$$

For a first-order deformation of φ_S, local lifts of P_a exist because ϕ is smooth along f_{P_a}. Their differences on overlaps form a Čech cocycle with values in $N_{P_a/S}$. Its class is δ_a; it vanishes precisely when the local lifts glue to a deformation of the section.

To pass to deformations over an unparameterized base, we quotient by PGL_2. This gives

$$T_{b_0} B \simeq H^0(\varphi_S^* T_{\mathcal{Y}}) \big/ d\varphi_S\big(H^0(T_{\mathbb{P}^1})\big)$$

The denominator has dimension three, since an infinitesimal source stabilizer would preserve the $12n$ distinct discriminant points and hence vanish. These reparametrizations lift through dg, so δ descends to $T_{b_0} B$.

We next compare this obstruction map with the period derivative. For a section $P = P_a$, consider

$$0 \longrightarrow \mathcal{O}_S \longrightarrow \mathcal{O}_S(P) \longrightarrow N_{P/S} \longrightarrow 0$$

Since $H^0(N_{P/S}) = 0$, we have $h^0(\mathcal{O}_S(P)) = 1$. Also $H^2(\mathcal{O}_S(P)) = 0$ by Serre duality: the divisor $K_S - P$ has intersection -1 with the nef fibre class, so it cannot be effective. Riemann–Roch gives

$$\chi(\mathcal{O}_S(P)) = n + \tfrac{1}{2} P \cdot (P - K_S) = n + \tfrac{1}{2}(-n - (n-2)) = 1$$

It follows that $H^1(\mathcal{O}_S(P)) = 0$. The connecting map, also called the divisor semiregularity map, is therefore an isomorphism

$$s_P : H^1(N_{P/S}) \xrightarrow{\sim} H^2(\mathcal{O}_S) \tag{27}$$

For the Cartier divisor $P \subset S$, the semiregularity map of Buchweitz–Flenner agrees with s_P: both are Serre dual to the restriction map $H^0(K_S) \to H^0(K_S|_P)$. The Kodaira–Spencer map associates to each tangent vector in $T_{b_0} B$ the corresponding first-order deformation class in $H^1(S, T_S)$. Compatibility with this map identifies the image of the section obstruction under s_P with the infinitesimal Hodge obstruction for $[P]$, taking values in $H^2(\mathcal{O}_S) \simeq H^{0,2}(S)$; see [BF00, Thms. 1.3, 4.5 and 5.5(1)–(2), §5.3]. Since $[O]$ and $[F]$ remain Hodge, this is also the obstruction for γ_a. Thus the composite

$$T_{b_0} B \xrightarrow{\delta} \bigoplus_{a=1}^{r} H^1(N_{P_a/S}) \xrightarrow{\oplus_a s_{P_a}} H^2(\mathcal{O}_S)^{\oplus r}$$

agrees with $d\Phi$, after applying Serre duality $H^2(\mathcal{O}_S) \simeq H^0(K_S)^\vee$ and evaluating in the chosen frame, up to a fixed non-zero scalar determined by the cycle-class and trace conventions. Since each s_{P_a} is an isomorphism, a first-order deformation lifts all the chosen sections if and only if their classes remain of type $(1,1)$ to first order.

The rank hypothesis now implies that δ is surjective. By (26), $H^1(E) = 0$. For maps from the fixed curve $\mathbb{P}^1$ to a smooth Deligne–Mumford stack, the tangent space is $H^0(E)$ and an obstruction space is $H^1(E)$. Indeed, local lifts across a square-zero extension differ by sections of E, and their obstruction to gluing is the resulting Čech class in $H^1(E)$. Thus the mapping stack is smooth at g. Using (25), its dimension is

$$h^0(E) = \chi(E) = (10n+1) + r(1-n)$$

Quotienting by PGL_2 subtracts 3, proving (24). In particular, for $r = 1$ under the rank hypothesis, the actual dimension equals the expected dimension $9n-1$, independently of d_1.

It remains to identify the germs. Since ρ is representable, the analytification of $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$, after base change to B, is an analytic space. Let H be a sufficiently small smooth analytic neighbourhood of the chosen lift h_0 in this base change.

The divisors P_a preserve their marked integral classes in this family. Thus the period functions Φ vanish on H, and, since H is reduced, ρ factors holomorphically through $\Phi^{-1}(0)$. The obstruction sequence and the quotient by source reparametrizations give

$$\begin{aligned} T_{h_0}H &\simeq H^0(E)\big/ dg\big(H^0(T_{\mathbb{P}^1})\big) \\ &\xrightarrow{d\rho} \ker\big(d\Phi(b_0)\big) \subset T_{b_0}B \end{aligned}$$

where the displayed arrow is an isomorphism. There are no additional tangent directions from moving the sections on a fixed surface because $H^0(N_{P_a/S}) = 0$. Both germs are smooth of dimension $10n-2-r(n-1)$, so the analytic inverse function theorem identifies them. The resulting identification is compatible with the forgetful morphism:

$$\begin{array}{ccc} (H, h_0) & \xrightarrow{\ \sim\ } & (\Phi^{-1}(0), b_0) \\ & {\scriptstyle\rho}\searrow \quad \swarrow & \\ & (B, b_0) & \end{array}$$

Smoothness here is asserted for $\mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}}$ and its local charts. ∎

The sections constructed in Theorem 3.11 satisfy the transversality hypothesis of the preceding proposition. We therefore obtain the following moduli-theoretic interpretation of the realization theorem.

Corollary 3.24. *Let $n \geq 2$ and $r \geq 1$ be integers satisfying $r(n-1) \leq 10n-2$. For every $[S] \in \mathcal{W}^{\mathrm{sb}}_n$, the surfaces and sections with indivisible Shioda classes of Theorem 3.11 give points $(S_m, P_{1,m}, \dots, P_{r,m}) \in \mathcal{S}^{\mathrm{I}_1}_{n,\mathbf{d}_m}$ with $d_{a,m} = \widehat{h}(P_{a,m}) \to \infty$ for every a. At each such point, the stack is smooth of dimension $10n-2-r(n-1)$, and its*

germ on the chosen marked chart is the simultaneous fixed-class Hodge locus germ. Moreover,

$$\deg f_{P_{a,m}}^{*}\mathcal{L}_{1,2} = n$$
$$\deg(\pi \circ f_{P_{a,m}})^{*}\mathcal{O}(1) = d_{a,m}$$
$$(P_{a,m} \cdot O_m) = d_{a,m} - n$$

Every analytic neighbourhood of every point of $\mathcal{W}_{\min,n}$ contains infinitely many pairwise non-isomorphic surfaces with these smooth section-moduli germs. For any prescribed $H > 0$, the surfaces and representatives of their germs may be chosen so that every non-zero Mordell–Weil section on every parametrized surface has canonical height greater than H.

Proof. Theorem 3.11 supplies independent sections with indivisible Shioda classes with heights tending to infinity and full-rank period derivatives. Proposition 3.23 gives the smoothness and the identification of germs, while Proposition 3.22 gives the degree formulas.

The final assertion follows from Corollary 3.13, using Proposition 3.23 to identify the resulting fixed-class Hodge germs with the section-moduli germs. ∎

We now combine the local construction, the height bounds, and the section-moduli comparison to complete the proofs of the main realization theorems.

Proof of Theorems 1.2 and 1.3. The construction in the proof of Theorem 3.11 takes place in a single marked local deformation chart B centred at b_0 representing $[S]$. It gives points $b_m \to b_0$ representing pairwise non-isomorphic surfaces, independent sections with indivisible Shioda classes, divergence of the least non-zero section height, and independence of the simultaneous period differentials.

Proposition 3.22 identifies the Shioda classes with $[P_{a,m}]-[O_m]-d_{a,m}[F_m]$, where $d_{a,m} = \widehat{h}(P_{a,m})$. Proposition 3.23 therefore gives the asserted smoothness, dimension, and identification of germs. Fixing the heights does not change these germs, since the heights are locally constant in families with chosen sections; restricting to the open substack $\mathcal{W}_n^{\mathrm{sb}}$ likewise does not change them.

Corollary 3.13 gives the assertions in arbitrary analytic neighbourhoods, including the uniform lower bounds for all non-zero section heights on representatives of the Hodge germs. After shrinking each representative, the germ comparison extends the r chosen sections uniquely over it. Their marked Shioda classes remain flat, so their height-pairing matrix is constant. Taking $r = 10$ gives dimension $10n-2-10(n-1) = 8$. ∎

Remark 3.25. The number of additional sections has a natural interpretation in the birational geometry of pointed genus one curves. The open locus in $\mathcal{X}^{(r)}$ parametrizing smooth elliptic curves with $O, P_1, \ldots, P_r$ pairwise distinct identifies with $\mathcal{M}_{1,r+1}$. Thus $\mathcal{X}^{(r)}$ is birational to $\overline{\mathcal{M}}_{1,r+1}$: the zero section contributes one marking. Writing κ for Kodaira dimension, the compactified coarse moduli spaces satisfy

$$\kappa\big(\overline{M}_{1,r+1}\big) = \begin{cases} -\infty & 1 \le r \le 9 \\ 0 & r = 10 \\ 1 & r \ge 11 \end{cases}$$

by [BF06, §4, Thm. 3 and the preceding discussion].

The same threshold appears in the canonical bundle and expected deformation dimension of our fibre-product target. Let $q\colon \mathcal{X}^{(r)} \to \mathcal{Y}$ be the common projection, and let $U \subset \mathcal{X}^{(r)}$ be the open locus where each coordinate lies in the relative smooth locus of ϕ. Since $\omega_{\mathcal{Y}} \simeq \lambda^{\otimes(-10)}$ and $\omega_{\mathcal{X}} \simeq \phi^*\lambda^{\otimes(-9)}$, we have

$$\omega_{\mathcal{X}/\mathcal{Y}} \simeq \phi^*\lambda \qquad \omega_U \simeq (q|_U)^*\lambda^{\otimes(r-10)}$$

All section maps above land in U, including at collisions. For $g\colon \mathbb{P}^1 \to U$ with $\deg(q\circ g)^*\lambda = n$, Riemann–Roch gives the expected dimension modulo PGL_2:

$$\chi(g^*T_U) - 3 = (10-r)n + r - 2 = 10n - 2 - r(n-1)$$

This is the actual local dimension at transverse tuples by Proposition 3.23. For $r = 10$, the canonical bundle of U is trivial and the expected dimension is $8 = \dim U - 3$, independently of n. For $r = 11$, the expected dimension is $9 - n$, so positive-dimensional families at height $n \geq 9$ have excess dimension. More generally, non-negativity of the expected dimension $10n-2-r(n-1)$ requires $r(n-1) \leq 10n-2$, which stabilizes at $r \leq 10$ for $n \geq 10$; beyond this range, every realization is non-transverse.

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